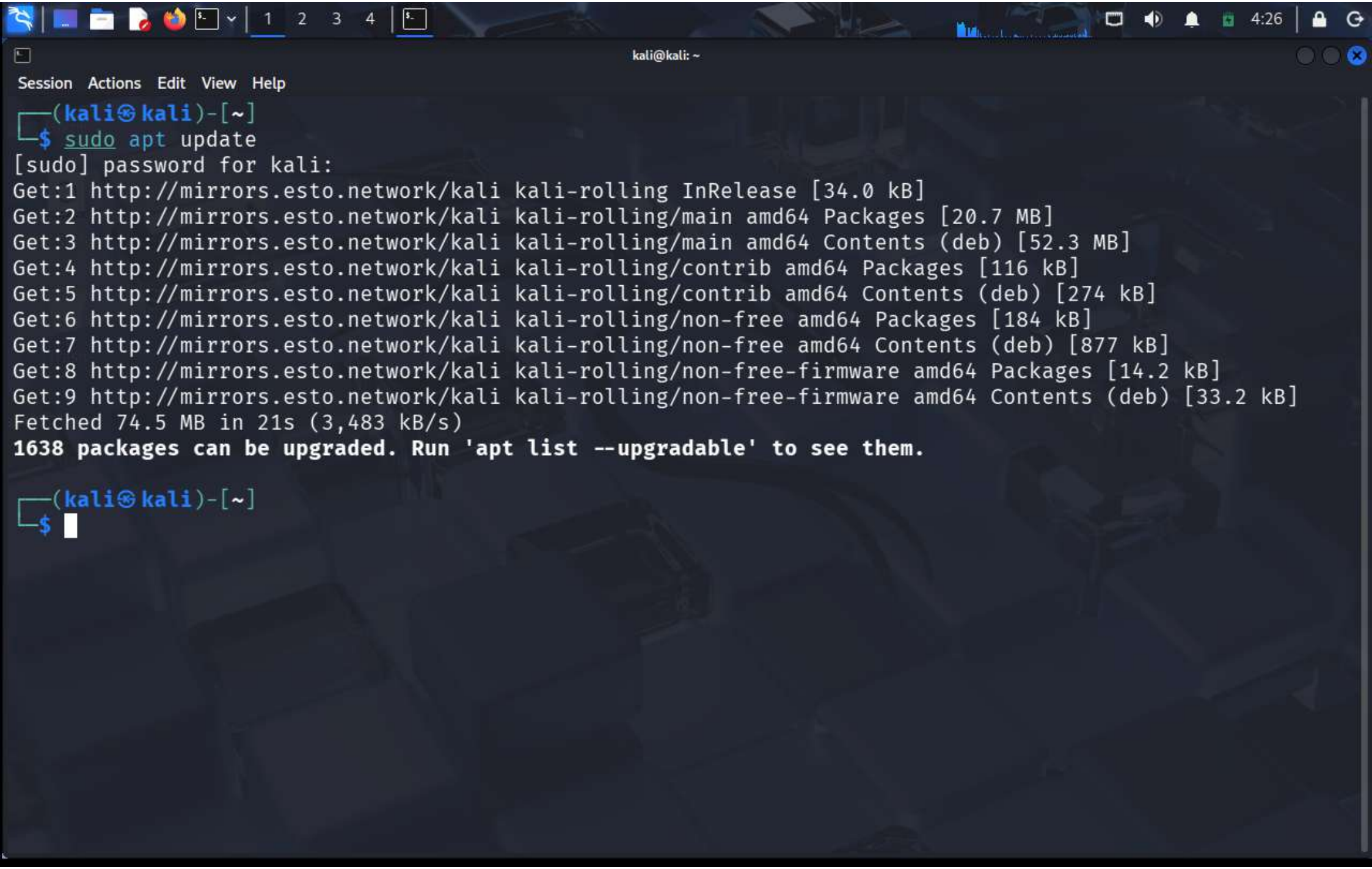

1 2 3 4
4:26
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo apt update
[sudo] password for kali:
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [20.7 MB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [52.3 MB]
Get:4 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [274 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [184 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [877 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [14.2 kB]
Get:9 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents (deb) [33.2 kB]
Fetched 74.5 MB in 21s (3,483 kB/s)
1638 packages can be upgraded. Run 'apt list --upgradable' to see them.
┌──(kali㉿kali)-[~]
└─$

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install -y docker.io
Installing:
  docker.io

Installing dependencies:
  containerd  docker-buildx  libcompel1        python3-pycriu  tini-static
  criu        docker-cli     python3-protobuf  runc

Suggested packages:
  containernetworking-plugins  btrfs-progs  rinse        xfsprogs  | zfsutils-linux
  docker-doc                   debootstrap  rootlesskit  zfs-fuse

Recommended packages:
  needrestart

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 1638
  Download size: 86.1 MB
  Space needed: 359 MB / 63.1 GB available

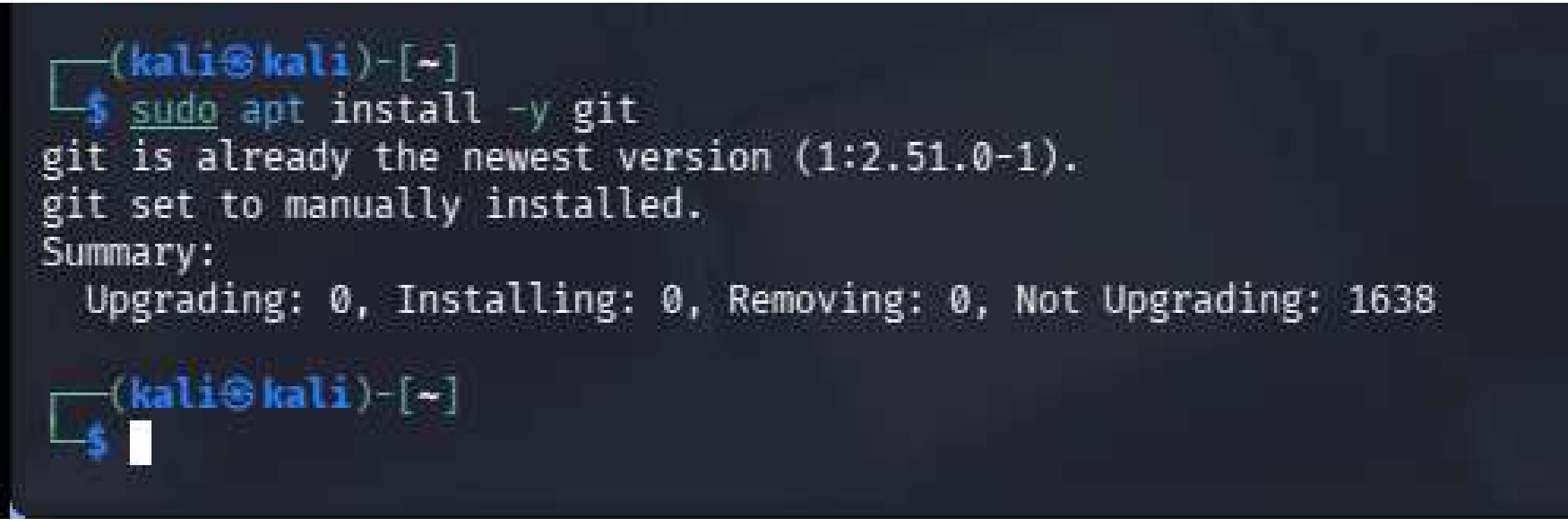
Get:1 http://http.kali.org/kali kali-rolling/main amd64 runc amd64 1.3.3+ds1-2 [6,686 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 containerd amd64 1.7.24~ds1-10 [33.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 tini-static amd64 0.19.0-6 [277 kB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 docker.io amd64 27.5.1+dfsg4-1 [23.2 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 libcompel1 amd64 4.2-1 [64.2 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 criu amd64 4.2-1 [557 kB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 docker-buildx amd64 0.19.3+ds1-4 [13.9 MB]
Get:8 http://http.kali.org/kali kali-rolling/main amd64 docker-cli amd64 27.5.1+dfsg4-1 [7,650 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 python3-protobuf amd64 3.21.12-15 [158 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 python3-pycriu all 4.2-1 [44.3 kB]
Fetched 86.1 MB in 11s (7,698 kB/s)
Selecting previously unselected package runc.
(Reading database ... 422308 files and directories currently installed.)
Preparing to unpack .../0-runc_1.3.3+ds1-2_amd64.deb ...
Unpacking runc (1.3.3+ds1-2) ...
Selecting previously unselected package containerd.
Preparing to unpack .../1-containerd_1.7.24~ds1-10_amd64.deb ...
Unpacking containerd (1.7.24~ds1-10) ...
Selecting previously unselected package tini-static.
Preparing to unpack .../2-tini-static_0.19.0-6_amd64.deb ...
Unpacking tini-static (0.19.0-6) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../3-docker.io_27.5.1+dfsg4-1_amd64.deb ...
Unpacking docker.io (27.5.1+dfsg4-1) ...
```

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install -y docker-compose
Installing:
  docker-compose

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1638
  Download size: 13.5 MB
  Space needed: 68.1 MB / 62.8 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 docker-compose amd64 2.32.4-3 [13.5 MB]
Fetched 13.5 MB in 12s (1,157 kB/s)
Selecting previously unselected package docker-compose.
(Reading database ... 422844 files and directories currently installed.)
Preparing to unpack .../docker-compose_2.32.4-3_amd64.deb ...
Unpacking docker-compose (2.32.4-3) ...
Setting up docker-compose (2.32.4-3) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for man-db (2.13.1-1) ...

┌──(kali㉿kali)-[~]
└─$
```


┌──(kali㉿kali)-[~]
└─$ sudo apt install -y git
git is already the newest version (1:2.51.0-1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1638
┌──(kali㉿kali)-[~]
└─$

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install -y git
git is already the newest version (1:2.51.0-1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1638

┌──(kali㉿kali)-[~]
└─$ sudo apt install -y qemu-system-x86
Upgrading:
  libc-bin  libc-dev-bin  libc-l10n  libc6  libc6-dev  libc6-i386  locales

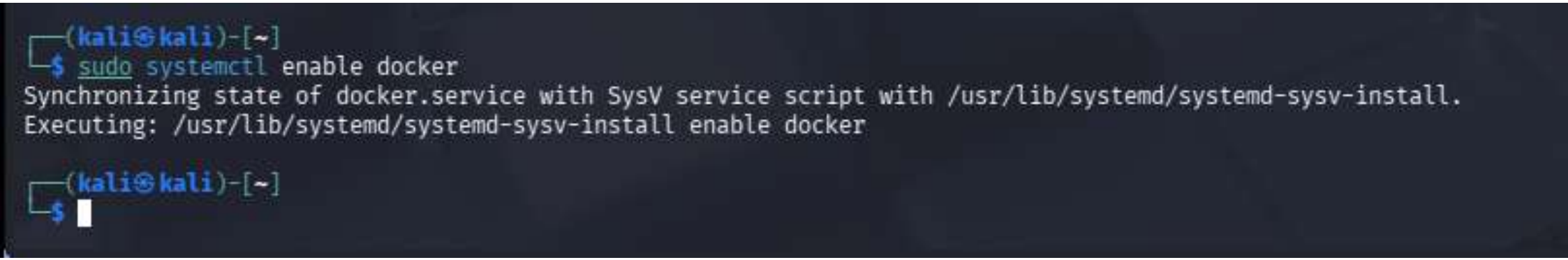
Installing:
  qemu-system-x86

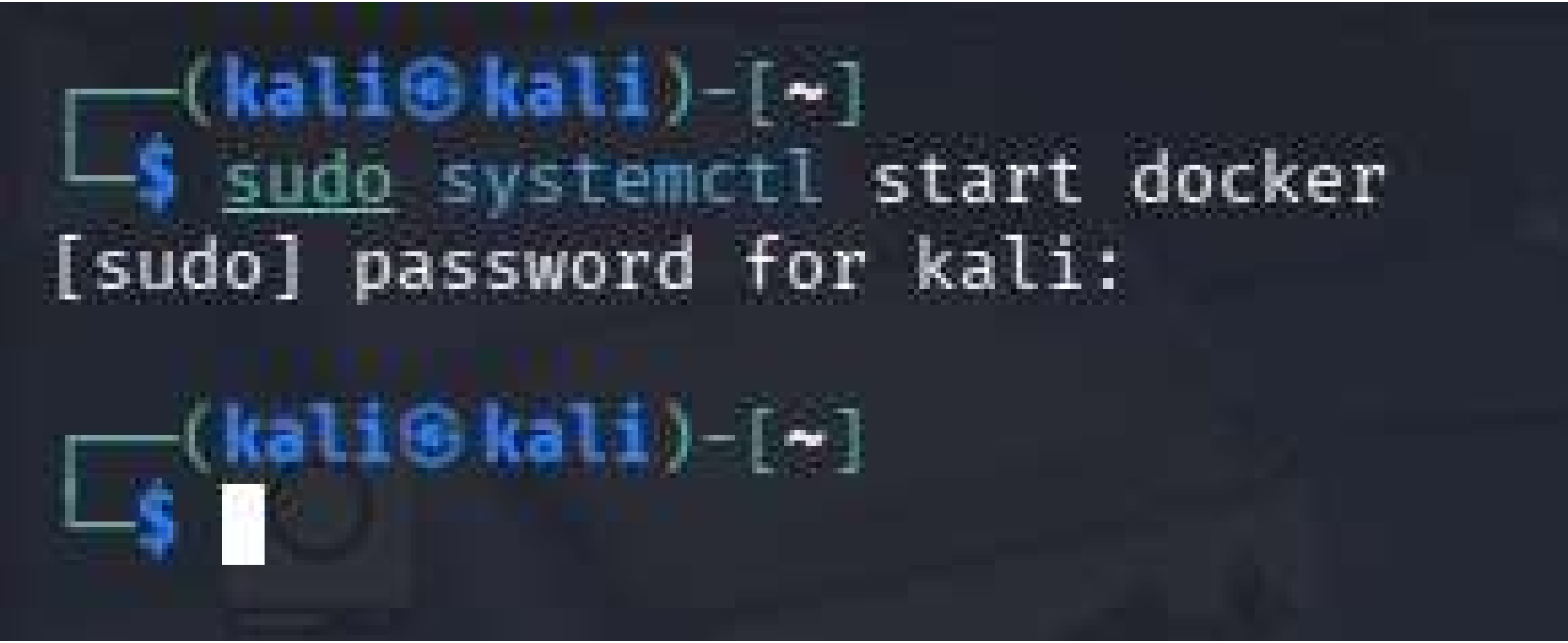
Installing dependencies:
  ibverbs-providers        libc-gconv-modules-extra  libgfapi0     libiscsi7  librdmacm1t64          libvirglrenderer1  qemu-system-gui             virgl-server
  ipxe-qemu                libcacard0                libgfrpc0     libndctl6  libslirp0              ovmf               qemu-system-modules-opengl
  libblkio1                libdaxctl1                libgfxdr0     libpmem1   libspice-server1       qemu-block-extra   qemu-system-modules-spice
  libboost-iostreams1.83.0 libexecs1                 libglusterfs0 librados2  libusbredirparser1t64  qemu-system-common qemu-utils
  libboost-thread1.83.0    libfdt1                   libibverbs1   librbd1    libvdeplug2t64         qemu-system-data   seabios

Suggested packages:
  gstreamer1.0-plugins-ugly  vde2  passt

Summary:
  Upgrading: 7, Installing: 37, Removing: 0, Not Upgrading: 1631
  Download size: 46.3 MB
  Space needed: 175 MB / 62.7 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 libc-gconv-modules-extra amd64 2.42-5 [1,127 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libc-l10n all 2.42-5 [749 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 locales all 2.42-5 [3,927 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libc6 amd64 2.42-5 [1,888 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 libc6-i386 amd64 2.42-5 [2,557 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 libc-dev-bin amd64 2.42-5 [60.3 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libc6-dev amd64 2.42-5 [2,091 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 libibverbs1 amd64 61.0-2 [64.2 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 ibverbs-providers amd64 61.0-2 [381 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/main amd64 libc-bin amd64 2.42-5 [674 kB]
Get:11 http://http.kali.org/kali kali-rolling/main amd64 ipxe-qemu all 1.21.1+git-20250829.969ce2c55+dfsg-3 [1,137 kB]
Get:13 http://http.kali.org/kali kali-rolling/main amd64 libboost-iostreams1.83.0 amd64 1.83.0-5+b1 [254 kB]
Get:14 http://http.kali.org/kali kali-rolling/main amd64 libboost-thread1.83.0 amd64 1.83.0-5+b1 [270 kB]
Get:15 http://kali.download/kali kali-rolling/main amd64 libcacard0 amd64 1:2.8.1-2 [96.9 kB]
Get:16 http://kali.download/kali kali-rolling/main amd64 libdaxctl1 amd64 82-1 [22.6 kB]
Get:17 http://http.kali.org/kali kali-rolling/main amd64 libexecs1 amd64 1.4-2+b3 [7,832 B]
```


(kali㉿kali)-[~]
$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
(kali㉿kali)-[~]
$


(kali㉿kali)-[~]
$ sudo systemctl start docker
[sudo] password for kali:
(kali㉿kali)-[~]
$

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl start docker
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-02-25 04:35:19 EST; 1min 31s ago
 Invocation: 78e1185004ed4598b6ba5df7ca38f03e
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 798 (dockerd)
      Tasks: 11
     Memory: 111M (peak: 111.4M)
        CPU: 1.257s
     CGroup: /system.slice/docker.service
             └─798 /usr/sbin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Feb 25 04:35:17 kali dockerd[798]: time="2026-02-25T04:35:17.978211660-05:00" level=info msg="Starting up"
Feb 25 04:35:17 kali dockerd[798]: time="2026-02-25T04:35:17.985470336-05:00" level=info msg="OTEL tracing is not configured, using no-op tracer provider"
Feb 25 04:35:18 kali dockerd[798]: time="2026-02-25T04:35:18.236956823-05:00" level=info msg="[graphdriver] using prior storage driver: overlay2"
Feb 25 04:35:18 kali dockerd[798]: time="2026-02-25T04:35:18.241807447-05:00" level=info msg="Loading containers: start."
Feb 25 04:35:19 kali dockerd[798]: time="2026-02-25T04:35:19.135415804-05:00" level=info msg="Default bridge (docker0) is assigned with an IP address 172.>
Feb 25 04:35:19 kali dockerd[798]: time="2026-02-25T04:35:19.314904705-05:00" level=info msg="Loading containers: done."
Feb 25 04:35:19 kali dockerd[798]: time="2026-02-25T04:35:19.337506392-05:00" level=info msg="Docker daemon" commit=61416484 containerd-snapshotter=false >
Feb 25 04:35:19 kali dockerd[798]: time="2026-02-25T04:35:19.337947597-05:00" level=info msg="Daemon has completed initialization"
Feb 25 04:35:19 kali dockerd[798]: time="2026-02-25T04:35:19.362863058-05:00" level=info msg="API listen on /run/docker.sock"
Feb 25 04:35:19 kali systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-23/23 (END)
```

```
┌──(kali㉿kali)-[~]
└─$ git clone -b 3.7.0 https://github.com/mendersoftware/integration.git mender-server
Cloning into 'mender-server' ...
remote: Enumerating objects: 26440, done.
remote: Counting objects: 100% (454/454), done.
remote: Compressing objects: 100% (209/209), done.
Receiving objects: 100% (26440/26440), 7.58 MiB | 106.00 KiB/s, done.
remote: Total 26440 (delta 372), reused 258 (delta 245), pack-reused 25986 (from 4)
Resolving deltas: 100% (18549/18549), done.
warning: refs/tags/3.7.0 53a99b8d82549d04f6297424ec20949f45161316 is not a commit!
Note: switching to '58265fa189f798eaa9e4003ad6cceb551fb4e9d0'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false


┌──(kali㉿kali)-[~]
└─$
```


(kali㉿kali)-[~]
$ cd mender-server
(kali㉿kali)-[~/mender-server]
$

```
┌──(kali㉿kali)-[~/mender-server]
└─$ sudo apt install -y jq
Upgrading:
  libjq1

Installing:
  jq

Summary:
  Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 1630
  Download size: 253 kB
  Space needed: 138 kB / 62.5 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libjq1 amd64 1.8.1-4+b1 [167 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 jq amd64 1.8.1-4+b1 [85.7 kB]
Fetched 253 kB in 1s (238 kB/s)
(Reading database ... 423649 files and directories currently installed.)
Preparing to unpack .../libjq1_1.8.1-4+b1_amd64.deb ...
Unpacking libjq1:amd64 (1.8.1-4+b1) over (1.8.1-4) ...
Selecting previously unselected package jq.
Preparing to unpack .../jq_1.8.1-4+b1_amd64.deb ...
Unpacking jq (1.8.1-4+b1) ...
Setting up libjq1:amd64 (1.8.1-4+b1) ...
Setting up jq (1.8.1-4+b1) ...
Processing triggers for libc-bin (2.42-5) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

┌──(kali㉿kali)-[~/mender-server]
└─$
```

```
┌──(kali㉿kali)-[~/mender-server]
└─$ sudo ./demo --client qemux86-64 up
-- enabling client container
WARN[0000] /home/kali/mender-server/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confus
ion
[+] Pulling 129/129
 ✔ mender-mongo Pulled                                                                                                                                 65.9s
 ✔ mender-inventory Pulled                                                                                                                             25.1s
 ✔ mender-nats Pulled                                                                                                                                  25.1s
 ✔ mender-deviceconnect Pulled                                                                                                                         43.6s
 ✔ mender-device-auth Pulled                                                                                                                          105.1s
 ✔ mender-deviceconfig Pulled                                                                                                                         109.1s
 ✔ mender-workflows-worker Pulled                                                                                                                      68.8s
 ✔ mender-iot-manager Pulled                                                                                                                           73.1s
 ✔ mender-create-artifact-worker Pulled                                                                                                                59.4s
 ✔ mender-api-gateway Pulled                                                                                                                           56.1s
 ✔ mender-deployments Pulled                                                                                                                          102.9s
 ✔ mender-useradm Pulled                                                                                                                               75.8s
 ✔ mender-gui Pulled                                                                                                                                   84.8s
 ✔ mender-workflows-server Pulled                                                                                                                      39.8s
```

```
Starting the Mender demo environment ...

Usage:  docker compose [OPTIONS] COMMAND

Define and run multi-container applications with Docker

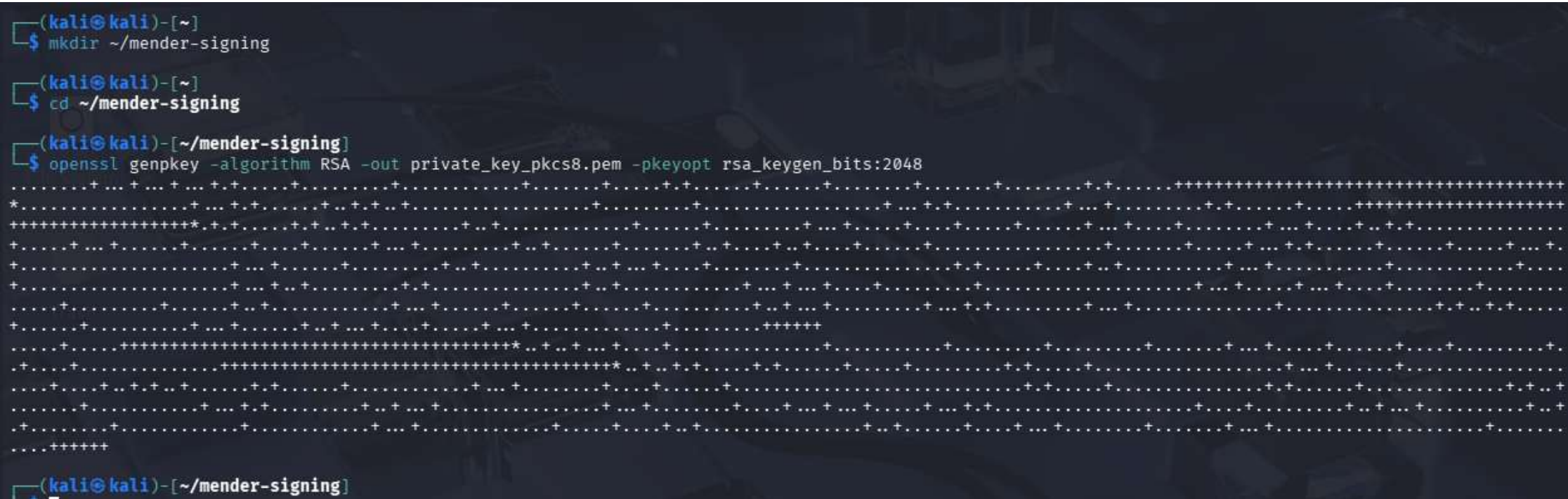
Options:
      --all-resources              Include all resources, even those not used by services
      --ansi string                Control when to print ANSI control characters ("never"|"always"|"auto") (default "auto")
      --compatibility              Run compose in backward compatibility mode
      --dry-run                    Execute command in dry run mode
      --env-file stringArray       Specify an alternate environment file
  -f, --file stringArray           Compose configuration files
      --parallel int               Control max parallelism, -1 for unlimited (default -1)
      --profile stringArray        Specify a profile to enable
      --progress string            Set type of progress output (auto, tty, plain, json, quiet) (default "auto")
      --project-directory string   Specify an alternate working directory
                                   (default: the path of the, first specified, Compose file)
  -p, --project-name string        Project name

Commands:
  attach      Attach local standard input, output, and error streams to a service's running container
  build       Build or rebuild services
  commit      Create a new image from a service container's changes
  config      Parse, resolve and render compose file in canonical format
  cp          Copy files/folders between a service container and the local filesystem
  create      Creates containers for a service
  down        Stop and remove containers, networks
  events      Receive real time events from containers
  exec        Execute a command in a running container
  export      Export a service container's filesystem as a tar archive
  images      List images used by the created containers
  kill        Force stop service containers
  logs        View output from containers
  ls          List running compose projects
  pause       Pause services
  port        Print the public port for a port binding
  ps          List containers
  pull        Pull service images
  push        Push service images
  restart     Restart service containers
  rm          Removes stopped service containers
```

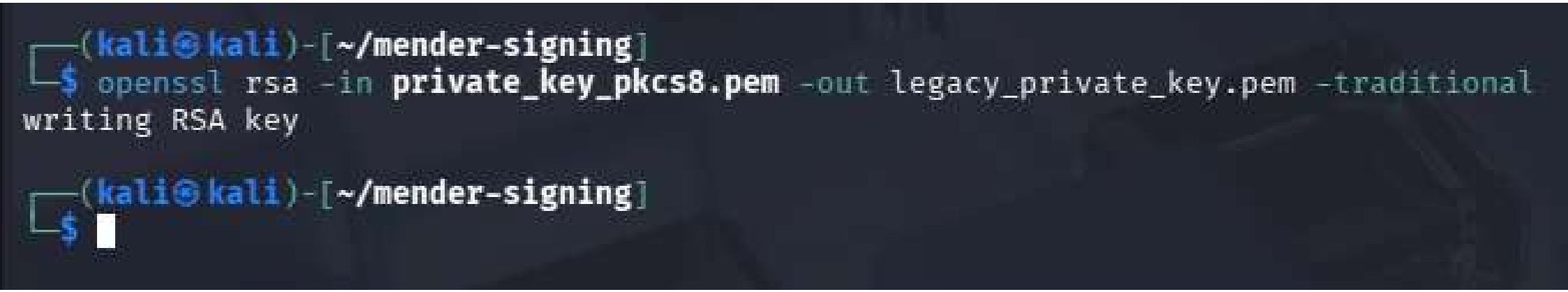
```
┌──(kali㉿kali)-[~]
└─$ mkdir ~/mender-signing

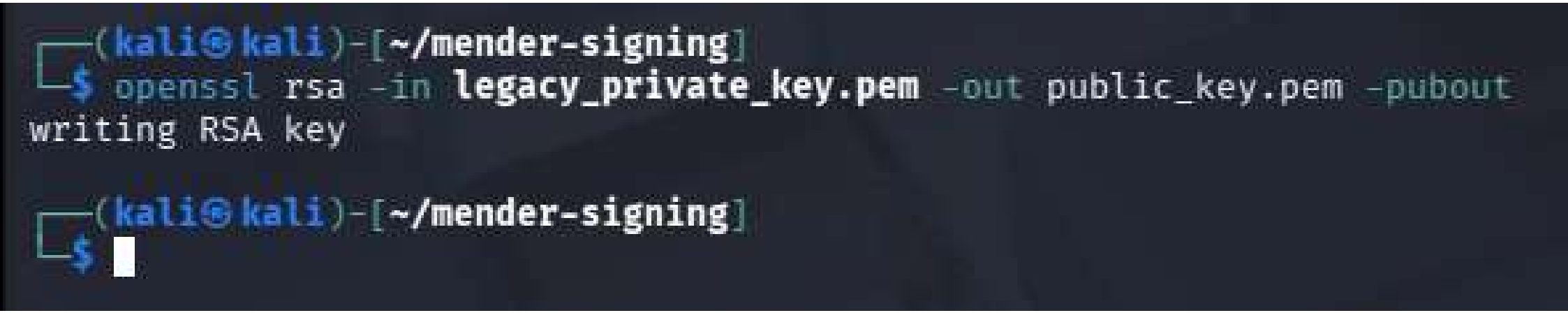
┌──(kali㉿kali)-[~]
└─$ cd ~/mender-signing

┌──(kali㉿kali)-[~/mender-signing]
└─$ openssl genpkey -algorithm RSA -out private_key_pkcs8.pem -pkeyopt rsa_keygen_bits:2048
```



```
┌──(kali㉿kali)-[~/mender-signing]
└─$
```


┌──(kali㉿kali)-[~/mender-signing]
└─$ openssl rsa -in private_key_pkcs8.pem -out legacy_private_key.pem -traditional
writing RSA key
┌──(kali㉿kali)-[~/mender-signing]
└─$


(kali㉿kali)-[~/mender-signing]
$ openssl rsa -in legacy_private_key.pem -out public_key.pem -pubout
writing RSA key
(kali㉿kali)-[~/mender-signing]
$

```
┌──(kali㉿kali)-[~/mender-signing]
└─$ echo "echo 'Spinnaker Analytics OTA Update Successful'" > payload.sh

┌──(kali㉿kali)-[~/mender-signing]
└─$ chmod +x payload.sh

┌──(kali㉿kali)-[~/mender-signing]
└─$ openssl dgst -sha256 -sign legacy_private_key.pem -out payload.sig payload.sh

┌──(kali㉿kali)-[~/mender-signing]
└─$
```

```
┌──(kali㉿kali)-[~/mender-signing]
└─$ wget https://downloads.mender.io/mender-artifact/master/linux/mender-artifact
--2026-02-25 04:50:56--  https://downloads.mender.io/mender-artifact/master/linux/mender-artifact
Resolving downloads.mender.io (downloads.mender.io)... 13.33.183.97, 13.33.183.55, 13.33.183.19, ...
Connecting to downloads.mender.io (downloads.mender.io)|13.33.183.97|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 29554040 (28M) [binary/octet-stream]
Saving to: 'mender-artifact'

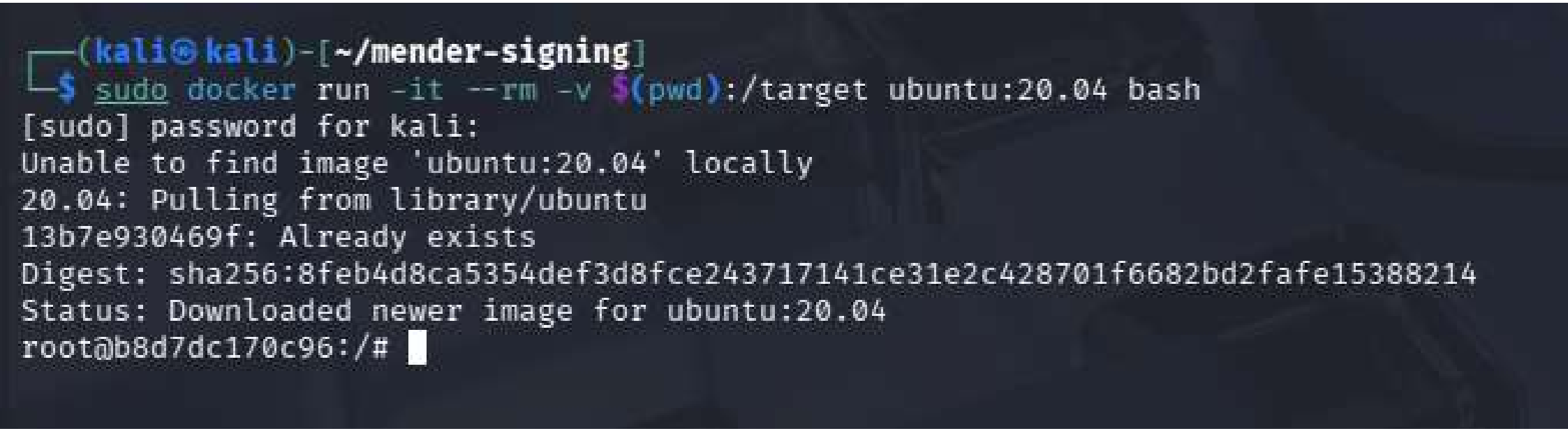
mender-artifact                     100%[═══════════════════════════════════════════════════════════════════════════⇒]  28.18M  7.20MB/s    in 4.7s

2026-02-25 04:51:01 (6.01 MB/s) - 'mender-artifact' saved [29554040/29554040]
```

```
└─$ sudo apt upgrade
The following packages were automatically installed and are no longer required:
  curlftpfs      libconfig-inifiles-perl  libmpeg2encpp-2.1-0t64  libpostproc58      libswscale8          python3-aiocache       python3-yaswfp
  libaudio2      libfuse2t64              libmplex2-2.1-0t64      librubberband2     libvdpau-va-gl1      python3-aiomcache      ruby-unf-ext
  libavfilter10  libgav1-1                libmupdf25.1            libsnmp40t64       mesa-vdpau-drivers   python3-fs             vdpau-driver-all
  libavformat61  libmjpegutils-2.1-0t64   libpocketsphinx3        libsphinxbase3t64  pocketsphinx-en-us   python3-wapiti-arsenic
Use 'sudo apt autoremove' to remove them.

Upgrading:
  7zip                          libfbclient2                  libqt6printsupport6           pkexec
  accountsservice               libfcgi-bin                   libqt6sql6                    plocate
  acl                           libfcgi0t64                   libqt6sql6-sqlite             plymouth
  adduser                       libfdisk1                     libqt6test6                   plymouth-label
  alsa-ucm-conf                 libffado2                     libqt6widgets6                pocl-opencl-icd
  amd64-microcode               libffi-dev                    libqt6xml6                    polkitd
  apache2                       libffi8                       libqtermwidget6-2             postgresql-18
  apache2-bin                   libfftw3-double3              libquadmath0                  postgresql-18-jit
  apache2-data                  libfftw3-single3              librabbitmq4                  postgresql-18-pg-gvm
  apache2-utils                 libfido2-1                    libraptor2-0                  postgresql-client-18
  apparmor                      libfile-desktopentry-perl     libraqm0                      powershell-empire
  apt                           libfile-mimeinfo-perl         librav1e0.8                   ppp
  apt-utils                     libfl-dev                     libraw1394-11                 procps
  arping                        libfl2                        libraw23t64                   proj-bin
  aspell                        libflashrom1                  libre2-11                     proj-data
  at-spi2-common                libflite1                     libreadline8t64               proxychains4
  at-spi2-core                  libfluidsynth3                librist4                      publicsuffix
  avahi-daemon                  libfmt10                      libroc0.4                     pv
  avahi-utils                   libfontconfig1                librpm-sequoia-1              pyinstaller
  axel                          libfontenc1                   librpm10                      pyqt6-dev-tools
  base-files                    libfreeaptx0                  librpmbuild10                 python-apt-common
  bind9-dnsutils                libfreerdp-client3-3          librpmio10                    python-babel-localedata
  bind9-host                    libfreerdp3-3                 librpmsign10                  python-tinycss2-common
  bind9-libs                    libfreetype6                  librsvg2-2                    python3
  binutils                      libfreexl1                    librsvg2-common               python3-aiodns
  binutils-common               libfribidi0                   librtlsdr0                    python3-aiofiles
  binutils-mingw-w64-i686       libfstrm0                     librtmp1                      python3-aiohttp
  binutils-mingw-w64-x86-64     libfsverity0                  librttopo1                    python3-alembic
  binutils-x86-64-linux-gnu     libftdi1-2                    libsamplerate0                python3-altgraph
  bsdextrautils                 libfuse2t64                   libsane1                      python3-apispec
  bsdutils                      libfuse3-4                    libsbc1                       python3-apt
  burpsuite                     libfyba0t64                   libsdl2-2.0-0                 python3-arc4
  busybox                       libgarcon-1-0                 libsdl2-classic               python3-asgiref
  bzip2                         libgarcon-gtk3-1-0            libsdl2-image-2.0-0           python3-asttokens
```


┌──(kali㉿kali)-[~/mender-signing]
└─$ wget https://downloads.mender.io/mender-artifact/master/linux/mender-artifact
--2026-02-25 05:47:28-- https://downloads.mender.io/mender-artifact/master/linux/mender-artifact
Resolving downloads.mender.io (downloads.mender.io)... 13.249.74.54, 13.249.74.65, 13.249.74.105, ...
Connecting to downloads.mender.io (downloads.mender.io)|13.249.74.54|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 29554040 (28M) [binary/octet-stream]
Saving to: 'mender-artifact'
mender-artifact 100%[==========>] 28.18M 7.58MB/s in 4.9s
2026-02-25 05:47:35 (5.74 MB/s) - 'mender-artifact' saved [29554040/29554040]
┌──(kali㉿kali)-[~/mender-signing]
└─$ chmod +x mender-artifact
┌──(kali㉿kali)-[~/mender-signing]
└─$


┌──(kali㉿kali)-[~/mender-signing]
└─$ sudo docker run -it --rm -v $(pwd):/target ubuntu:20.04 bash
[sudo] password for kali:
Unable to find image 'ubuntu:20.04' locally
20.04: Pulling from library/ubuntu
13b7e930469f: Already exists
Digest: sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214
Status: Downloaded newer image for ubuntu:20.04
root@b8d7dc170c96:/#

```
┌──(kali㉿kali)-[~/mender-signing]
└─$ sudo docker run -it --rm -v $(pwd):/target ubuntu:20.04 bash
[sudo] password for kali:
Unable to find image 'ubuntu:20.04' locally
20.04: Pulling from library/ubuntu
13b7e930469f: Already exists
Digest: sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214
Status: Downloaded newer image for ubuntu:20.04
root@b8d7dc170c96:/# apt-get update
Get:1 http://archive.ubuntu.com/ubuntu focal InRelease [265 kB]
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1308 kB]
Get:4 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:5 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [4432 kB]
Get:7 http://archive.ubuntu.com/ubuntu focal/restricted amd64 Packages [33.4 kB]
Get:8 http://archive.ubuntu.com/ubuntu focal/main amd64 Packages [1275 kB]
Get:9 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [4801 kB]
Get:10 http://archive.ubuntu.com/ubuntu focal/multiverse amd64 Packages [177 kB]
Get:11 http://archive.ubuntu.com/ubuntu focal/universe amd64 Packages [11.3 MB]
Get:12 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [33.1 kB]
Get:13 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [4920 kB]
Get:14 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1599 kB]
Get:15 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [4998 kB]
Get:16 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [36.8 kB]
Get:17 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [55.2 kB]
Get:18 http://archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [28.6 kB]
Fetched 35.7 MB in 9s (4134 kB/s)
Reading package lists... Done
root@b8d7dc170c96:/#
```

```
root@b8d7dc170c96:/# apt-get install -y libssl1.1
Reading package lists ... Done
Building dependency tree
Reading state information ... Done
The following NEW packages will be installed:
  libssl1.1
0 upgraded, 1 newly installed, 0 to remove and 5 not upgraded.
Need to get 1323 kB of archives.
After this operation, 4131 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 libssl1.1 amd64 1.1.1f-1ubuntu2.24 [1323 kB]
Fetched 1323 kB in 4s (304 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libssl1.1:amd64.
(Reading database ... 4124 files and directories currently installed.)
Preparing to unpack .../libssl1.1_1.1.1f-1ubuntu2.24_amd64.deb ...
Unpacking libssl1.1:amd64 (1.1.1f-1ubuntu2.24) ...
Setting up libssl1.1:amd64 (1.1.1f-1ubuntu2.24) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 76.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC contains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.30.0 /usr/local/share/perl/5.30.0 /usr/lib/x86_64-linux-gnu/perl5/5.30 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl/5.30 /usr/share/perl/5.30 /usr/local/lib/site_perl /usr/lib/x86_64-linux-gnu/perl-base) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
Processing triggers for libc-bin (2.31-0ubuntu9.17) ...
root@b8d7dc170c96:/#
```

```
root@b8d7dc170c96:/# cd /target
root@b8d7dc170c96:/target# ./mender-artifact write module-image -T script -n custom-signed-v1.0 -t qemux86-64 -o custom-signed-v1.mender -f payload.sh -k l
egacy_private_key.pem
Writing Artifact...
Version                 √
Manifest                √
Manifest signature      √
Header                  √
Payload
............................................................ - 100 %
root@b8d7dc170c96:/target#
```

```
root@b8d7dc170c96:/target# exit
exit

┌──(kali㉿kali)-[~/mender-signing]
└─$ ls -l custom-signed-v1.mender
-rw-r--r-- 1 root root 6144 Feb 25 05:52 custom-signed-v1.mender

┌──(kali㉿kali)-[~/mender-signing]
└─$
```

```
┌──(kali㉿kali)-[~/mender-server]
└─$ sudo ./demo up -d
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting the Mender demo environment ...
WARN[0000] /home/kali/mender-server/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusi
on
WARN[0000] /home/kali/mender-server/docker-compose.storage.minio.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid pot
ential confusion
WARN[0000] /home/kali/mender-server/docker-compose.demo.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
[+] Running 16/16
 ✔ Network menderserver_mender                          Created                                                                                    0.2s
 ✔ Container menderserver-minio-1                       Healthy                                                                                    6.5s
 ✔ Container menderserver-mender-nats-1                 Started                                                                                    0.9s
 ✔ Container menderserver-mender-mongo-1                Started                                                                                    1.0s
 ✔ Container menderserver-mender-gui-1                  Started                                                                                    0.9s
 ✔ Container menderserver-mender-iot-manager-1          Started                                                                                    2.8s
 ✔ Container menderserver-mender-create-artifact-worker-1  Started                                                                                 2.7s
 ✔ Container menderserver-mender-deviceconfig-1         Started                                                                                    2.0s
 ✔ Container menderserver-mender-deviceconnect-1        Started                                                                                    2.4s
 ✔ Container menderserver-mender-deployments-1          Started                                                                                    7.6s
 ✔ Container menderserver-mender-workflows-server-1     Started                                                                                    2.5s
 ✔ Container menderserver-mender-workflows-worker-1     Started                                                                                    2.6s
 ✔ Container menderserver-mender-inventory-1            Started                                                                                    2.6s
 ✔ Container menderserver-mender-useradm-1              Started                                                                                    2.4s
 ✔ Container menderserver-mender-device-auth-1          Started                                                                                    3.2s
 ✔ Container menderserver-mender-api-gateway-1          Started                                                                                    4.2s
Waiting for services to become ready ...
Creating a new user ...
WARN[0000] /home/kali/mender-server/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusi
on
WARN[0000] /home/kali/mender-server/docker-compose.storage.minio.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid pot
ential confusion
WARN[0000] /home/kali/mender-server/docker-compose.demo.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
time="2026-02-25T12:18:46Z" level=info msg="loading plan definitions from configuration" caller="main.doMain.func1@main.go:145"
****************************************
```

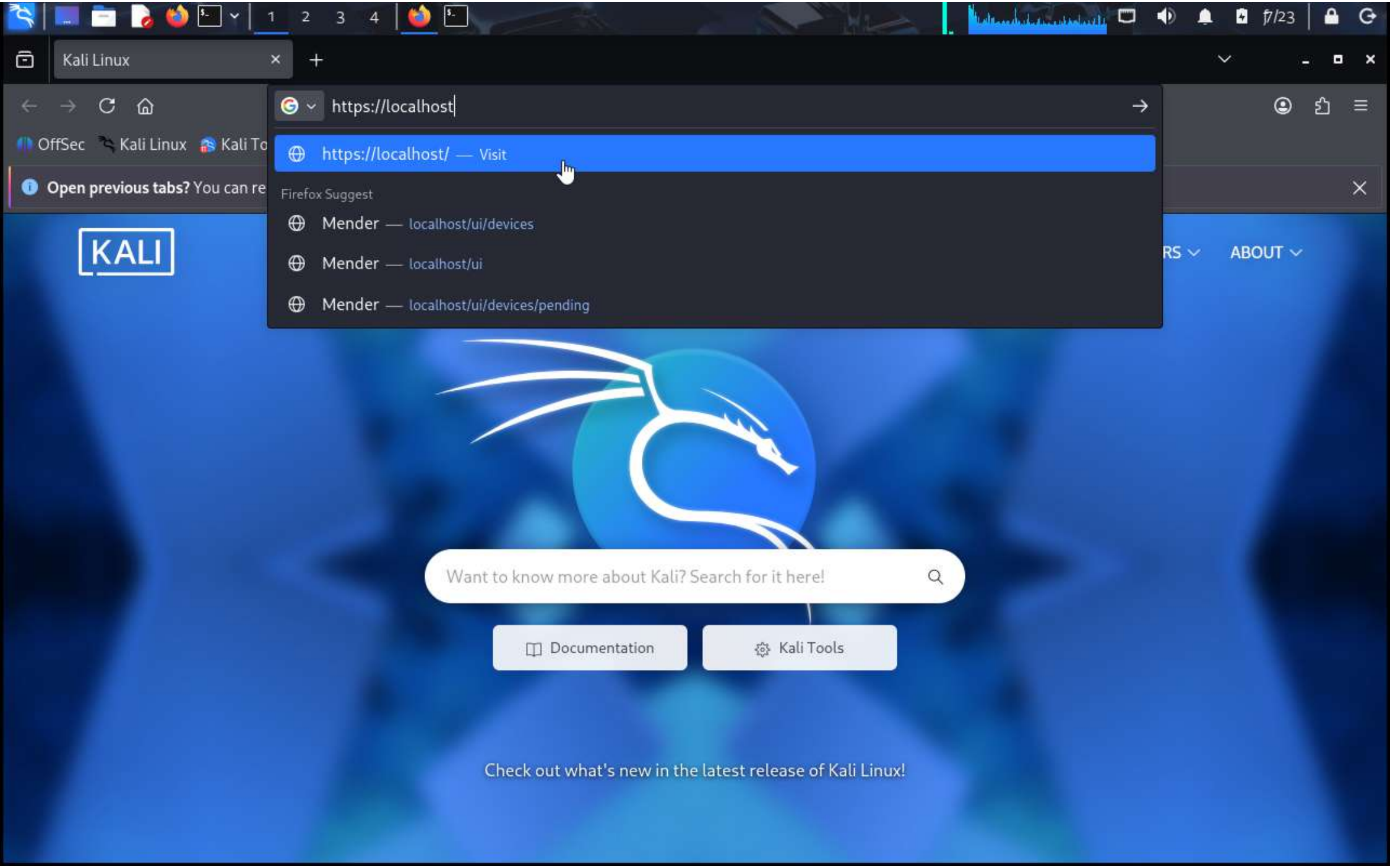
```
WARN[0000] /home/kali/mender-server/docker-compose.storage.minio.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid pot
ential confusion
WARN[0000] /home/kali/mender-server/docker-compose.demo.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
[+] Running 16/16
 ✔ Network menderserver_mender                                Created                                                                                0.2s
 ✔ Container menderserver-minio-1                             Healthy                                                                                6.5s
 ✔ Container menderserver-mender-nats-1                       Started                                                                                0.9s
 ✔ Container menderserver-mender-mongo-1                      Started                                                                                1.0s
 ✔ Container menderserver-mender-gui-1                        Started                                                                                0.9s
 ✔ Container menderserver-mender-iot-manager-1                Started                                                                                2.8s
 ✔ Container menderserver-mender-create-artifact-worker-1     Started                                                                                2.7s
 ✔ Container menderserver-mender-deviceconfig-1               Started                                                                                2.0s
 ✔ Container menderserver-mender-deviceconnect-1              Started                                                                                2.4s
 ✔ Container menderserver-mender-deployments-1                Started                                                                                7.6s
 ✔ Container menderserver-mender-workflows-server-1           Started                                                                                2.5s
 ✔ Container menderserver-mender-workflows-worker-1           Started                                                                                2.6s
 ✔ Container menderserver-mender-inventory-1                  Started                                                                                2.6s
 ✔ Container menderserver-mender-useradm-1                    Started                                                                                2.4s
 ✔ Container menderserver-mender-device-auth-1                Started                                                                                3.2s
 ✔ Container menderserver-mender-api-gateway-1                Started                                                                                4.2s
Waiting for services to become ready ...
Creating a new user ...
WARN[0000] /home/kali/mender-server/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusi
on
WARN[0000] /home/kali/mender-server/docker-compose.storage.minio.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid pot
ential confusion
WARN[0000] /home/kali/mender-server/docker-compose.demo.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
time="2026-02-25T12:18:46Z" level=info msg="loading plan definitions from configuration" caller="main.doMain.func1@main.go:145"
****************************************

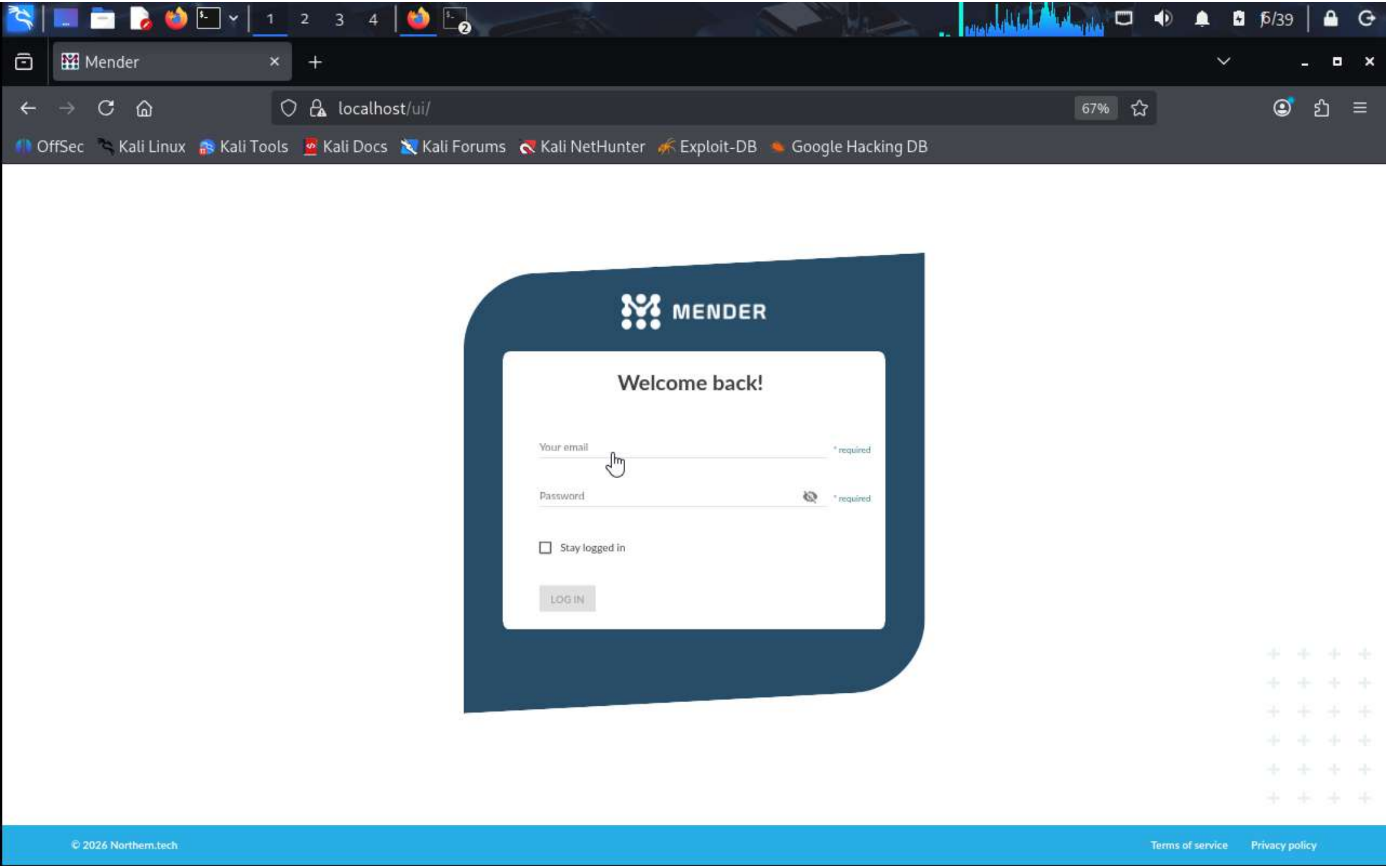
Username: mender-demo@example.com
Login password: 80F37CD1A61D

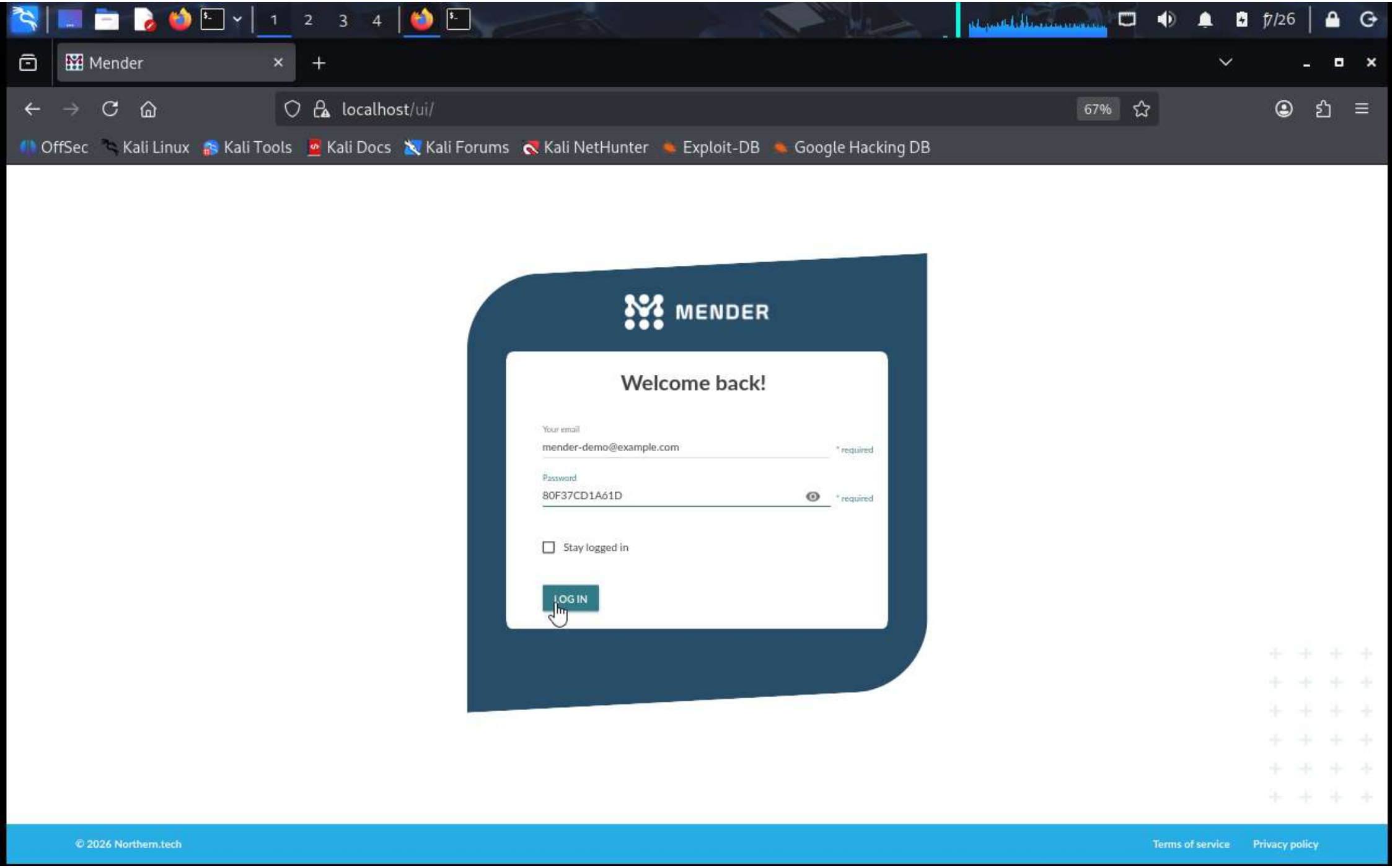
****************************************
Please keep the password available, it will not be cached by the login script.
Mender demo server ready and running in the background. Copy credentials above and log in at https://localhost
Press Enter to show the logs.
Press Ctrl-C to stop the backend and quit.
```

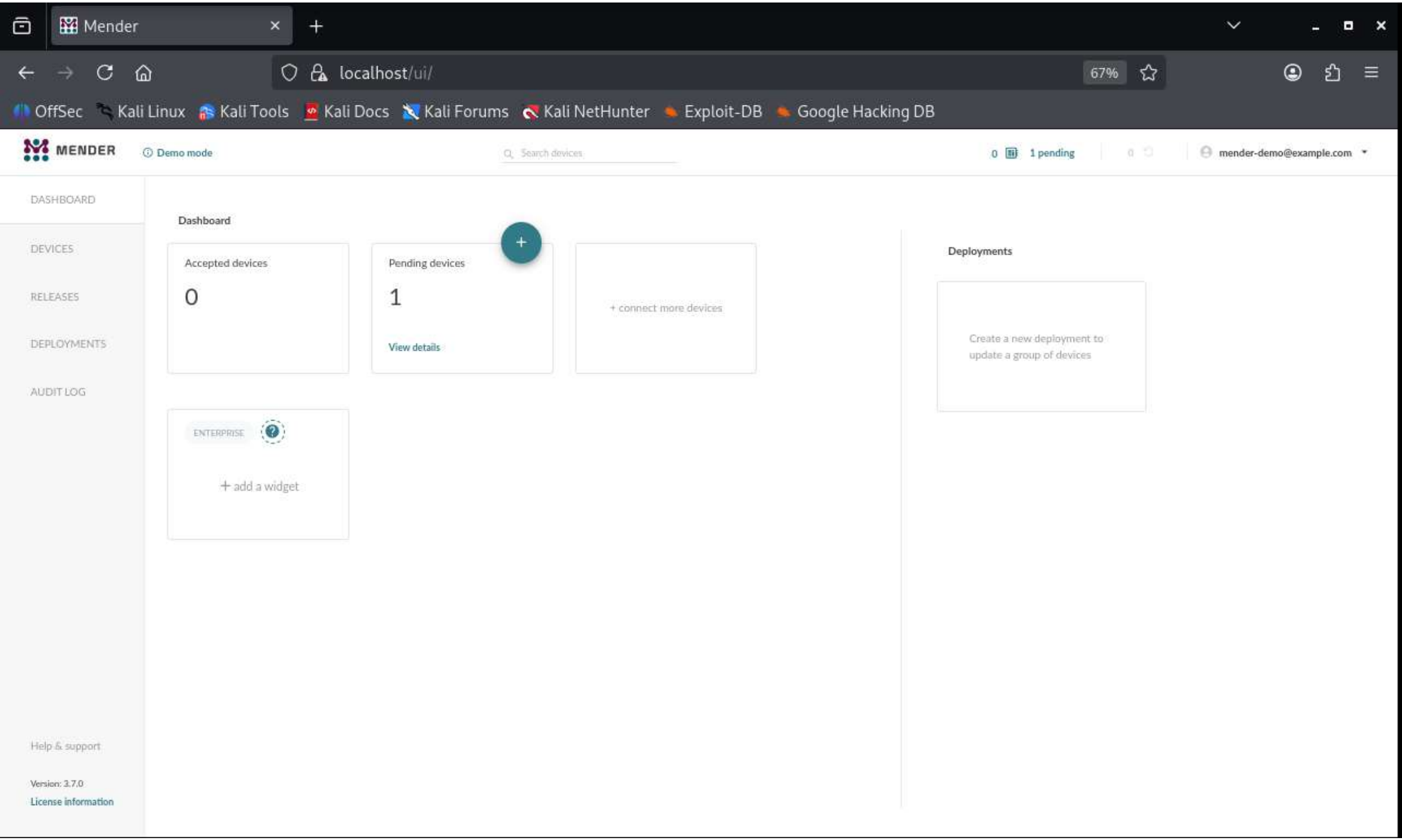
```
┌──(kali㉿kali)-[~/mender-server]
└─$ sudo docker compose -p menderserver -f docker-compose.client.yml up -d
WARN[0000] /home/kali/mender-server/docker-compose.client.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential
confusion
WARN[0000] Found orphan containers ([menderserver-mender-api-gateway-1 menderserver-mender-device-auth-1 menderserver-mender-create-artifact-worker-1 mender
server-mender-deviceconnect-1 menderserver-mender-deviceconfig-1 menderserver-mender-workflows-server-1 menderserver-mender-inventory-1 menderserver-mender-
useradm-1 menderserver-mender-iot-manager-1 menderserver-mender-deployments-1 menderserver-mender-workflows-worker-1 menderserver-mender-gui-1 menderserver-
mender-mongo-1 menderserver-minio-1 menderserver-mender-nats-1]) for this project. If you removed or renamed this service in your compose file, you can run
this command with the --remove-orphans flag to clean it up.
[+] Running 1/1
 ✔ Container menderserver-mender-client-1  Started                                                                                                  0.7s

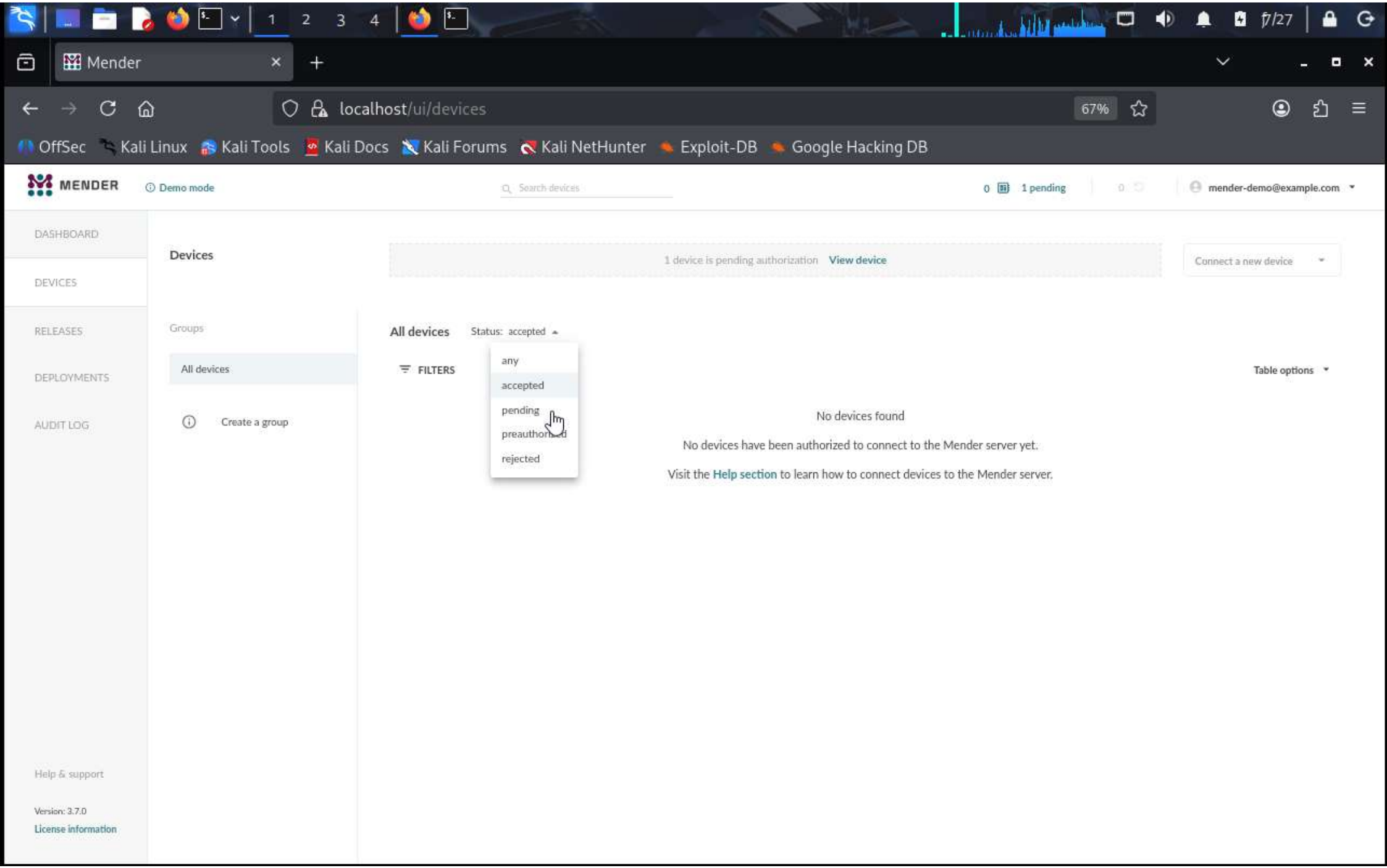
┌──(kali㉿kali)-[~/mender-server]
└─$
```

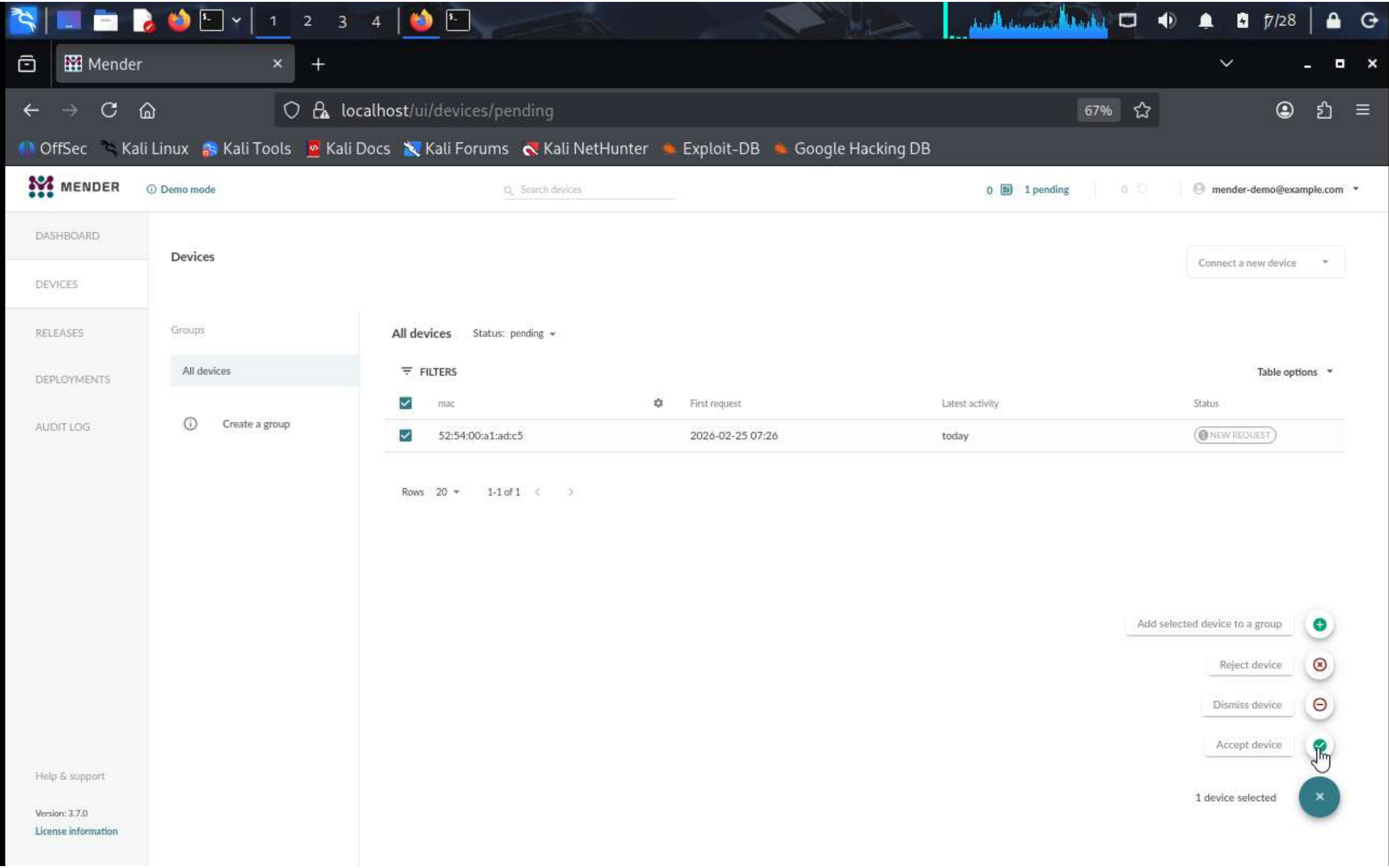

1 2 3 4
Kali Linux
https://localhost
https://localhost/ — Visit
Firefox Suggest
Mender — localhost/ui/devices
Mender — localhost/ui
Mender — localhost/ui/devices/pending
OffSec
Kali Linux
Open previous tabs? You can re
KALI
ABOUT
Want to know more about Kali? Search for it here!
Documentation
Kali Tools
Check out what's new in the latest release of Kali Linux!


Mender
localhost/ui/
67%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
MENDER
Welcome back!
Your email
* required
Password
* required
Stay logged in
LOG IN
© 2026 Northern.tech
Terms of service
Privacy policy

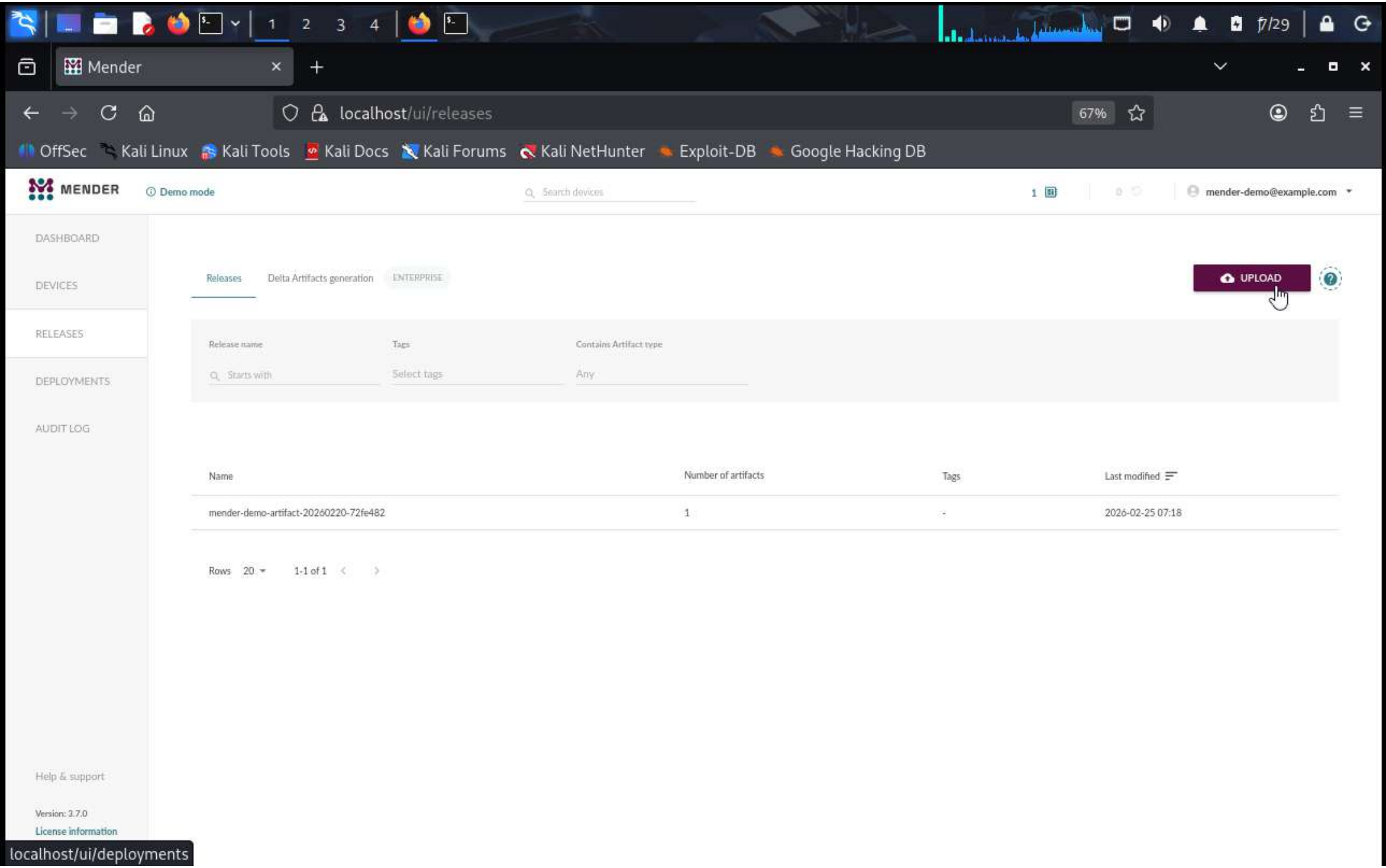

Mender
localhost/ui/
67%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
MENDER
Welcome back!
Your email
mender-demo@example.com
* required
Password
80F37CD1A61D
* required
Stay logged in
LOG IN
© 2026 Northern.tech
Terms of service
Privacy policy

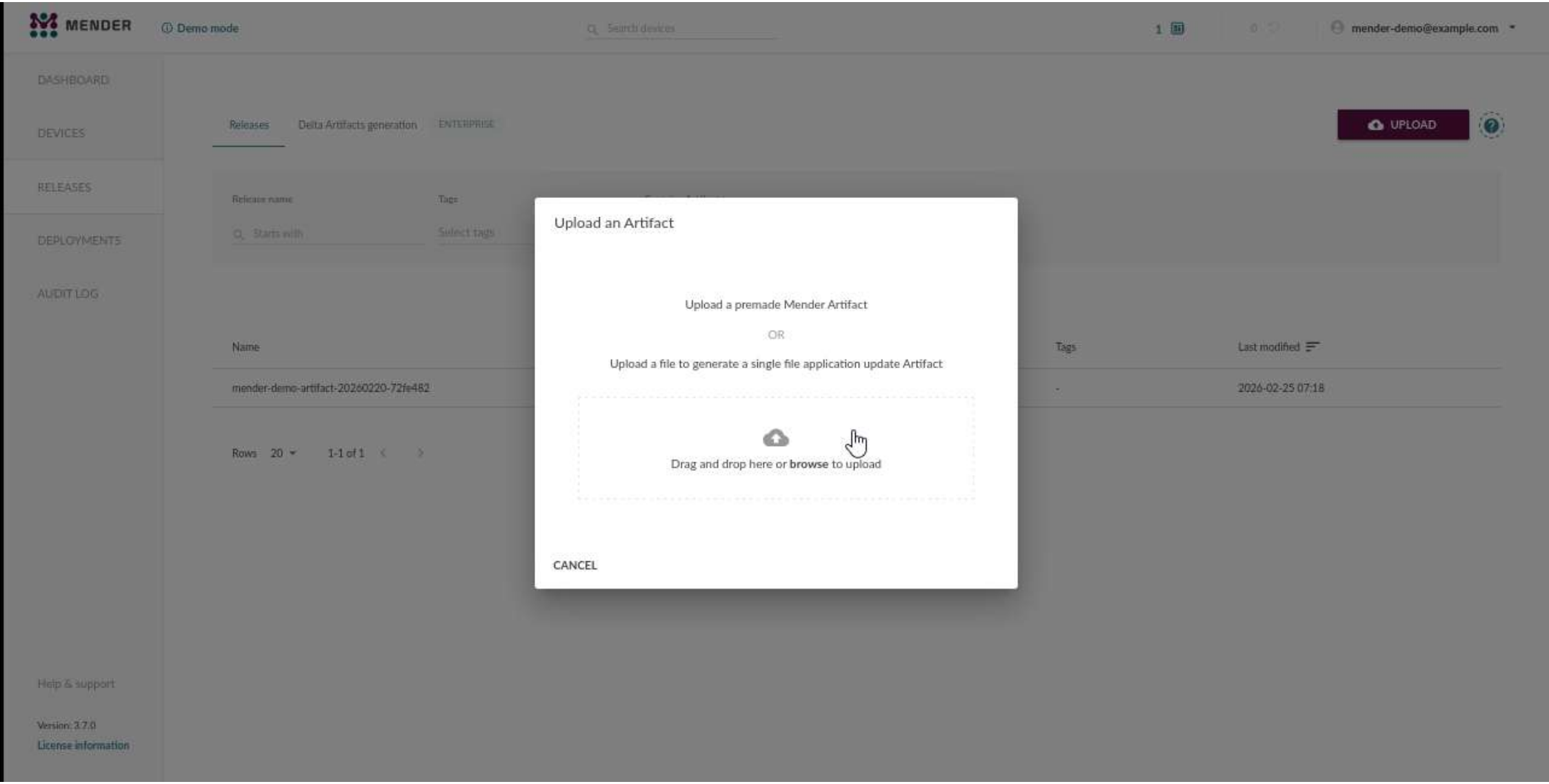

Mender
localhost/ui/
67%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
MENDER
Demo mode
Search devices
0
1 pending
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Dashboard
Accepted devices
0
Pending devices
1
View details
+ connect more devices
Deployments
Create a new deployment to update a group of devices
ENTERPRISE
+ add a widget
Help & support
Version: 3.7.0
License information

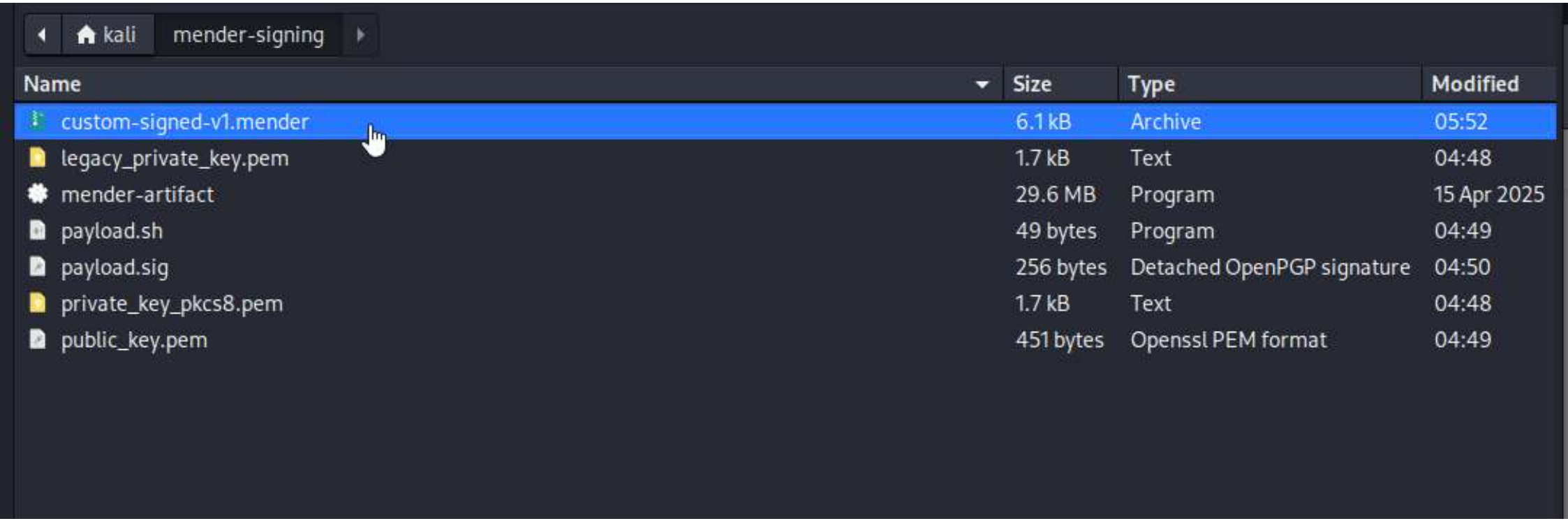

Mender
localhost/ui/devices
67%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
MENDER
Demo mode
Search devices
0
1 pending
0
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Help & support
Version: 3.7.0
License information
Devices
1 device is pending authorization
View device
Connect a new device
Groups
All devices
Create a group
All devices
Status: accepted
FILTERS
any
accepted
pending
preauthorized
rejected
Table options
No devices found
No devices have been authorized to connect to the Mender server yet.
Visit the Help section to learn how to connect devices to the Mender server.

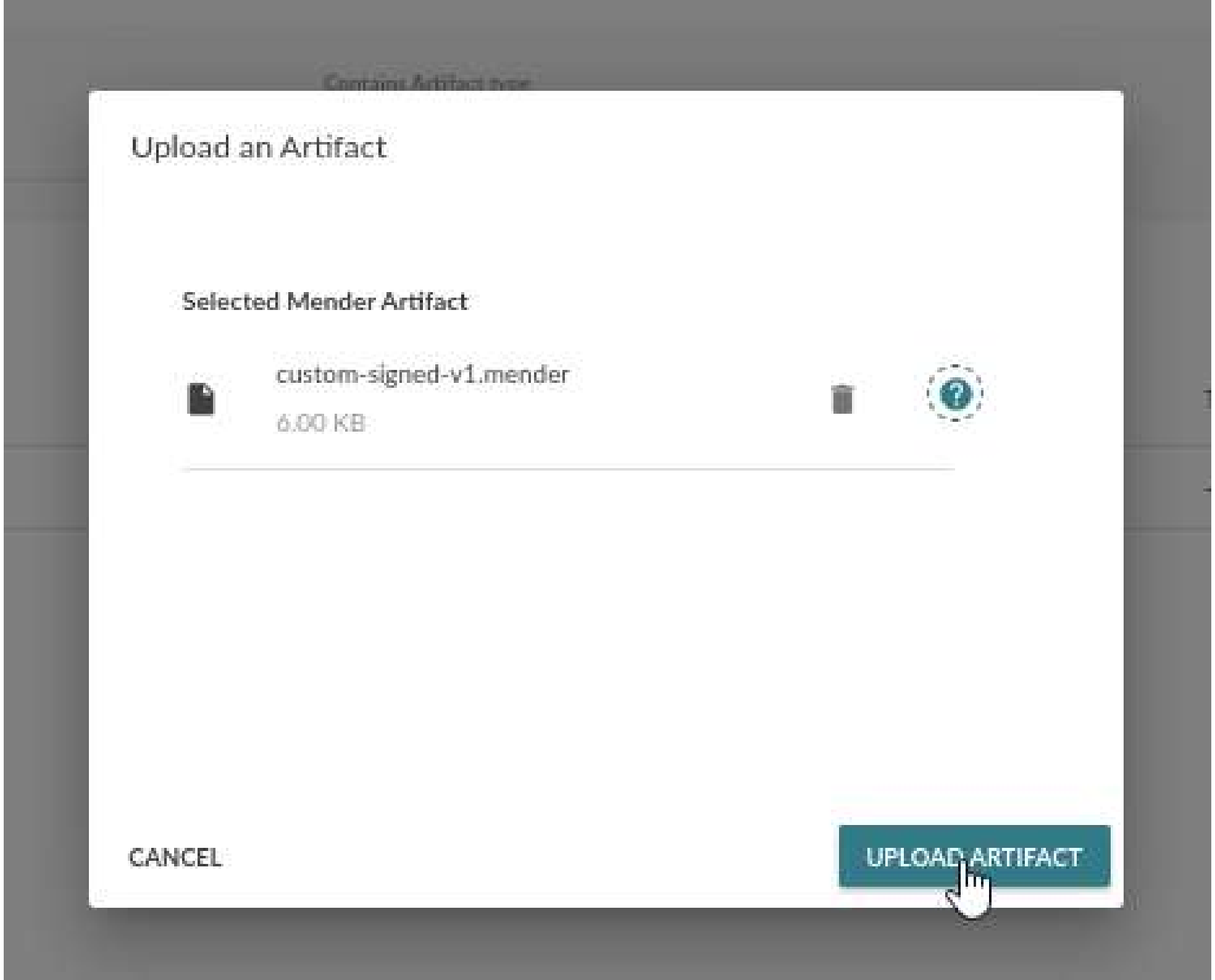

Mender
localhost/ui/devices/pending
67%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
MENDER
Demo mode
Search devices
0
1 pending
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Help & support
Version: 3.7.0
License information
Devices
Connect a new device
Groups
All devices
Create a group
All devices
Status: pending
FILTERS
Table options
mac
First request
Latest activity
Status
52:54:00:a1:ad:c5
2026-02-25 07:26
today
NEW REQUEST
Rows 20
1-1 of 1
Add selected device to a group
Reject device
Dismiss device
Accept device
1 device selected

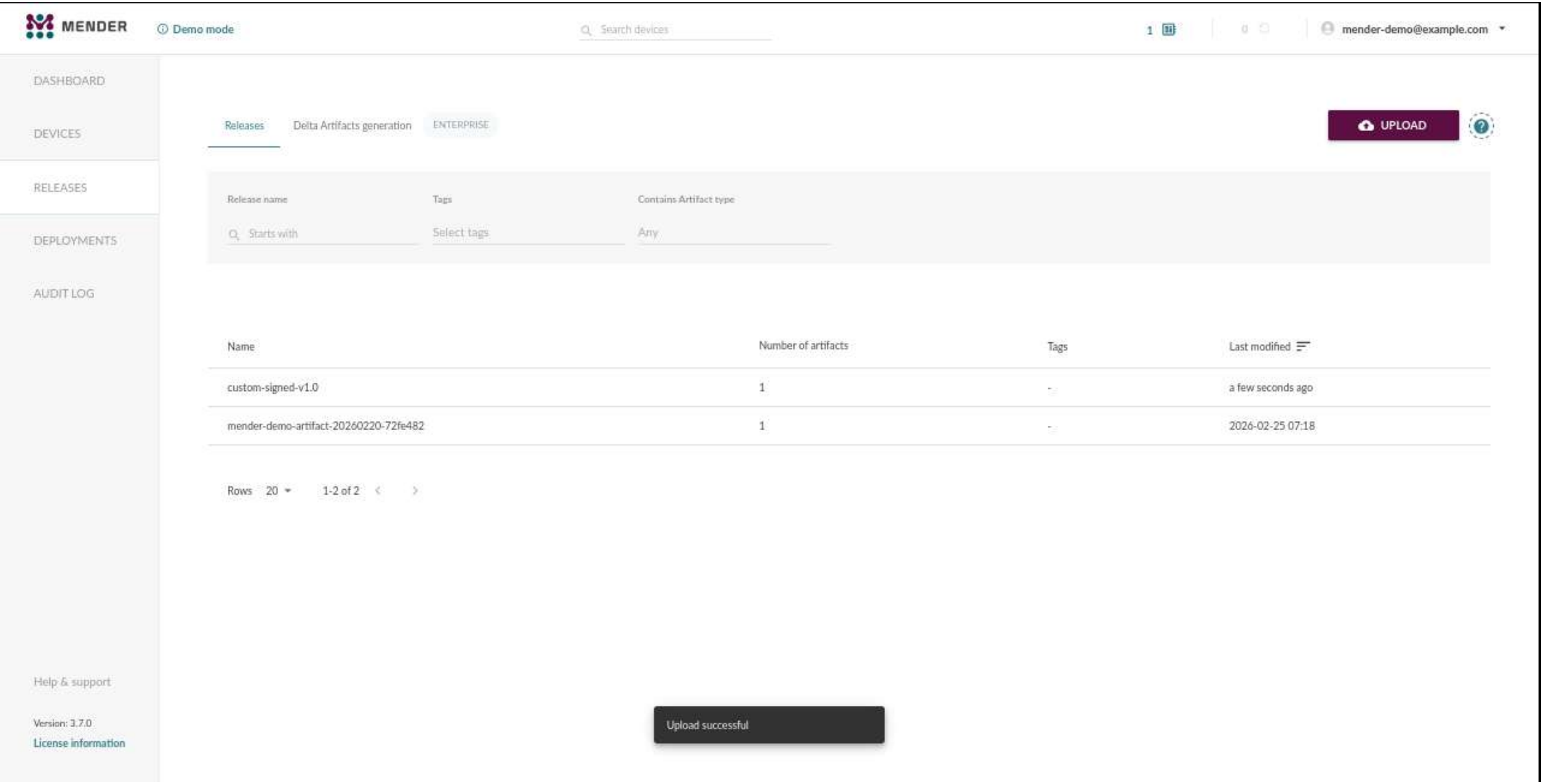

1 device was updated successfully.

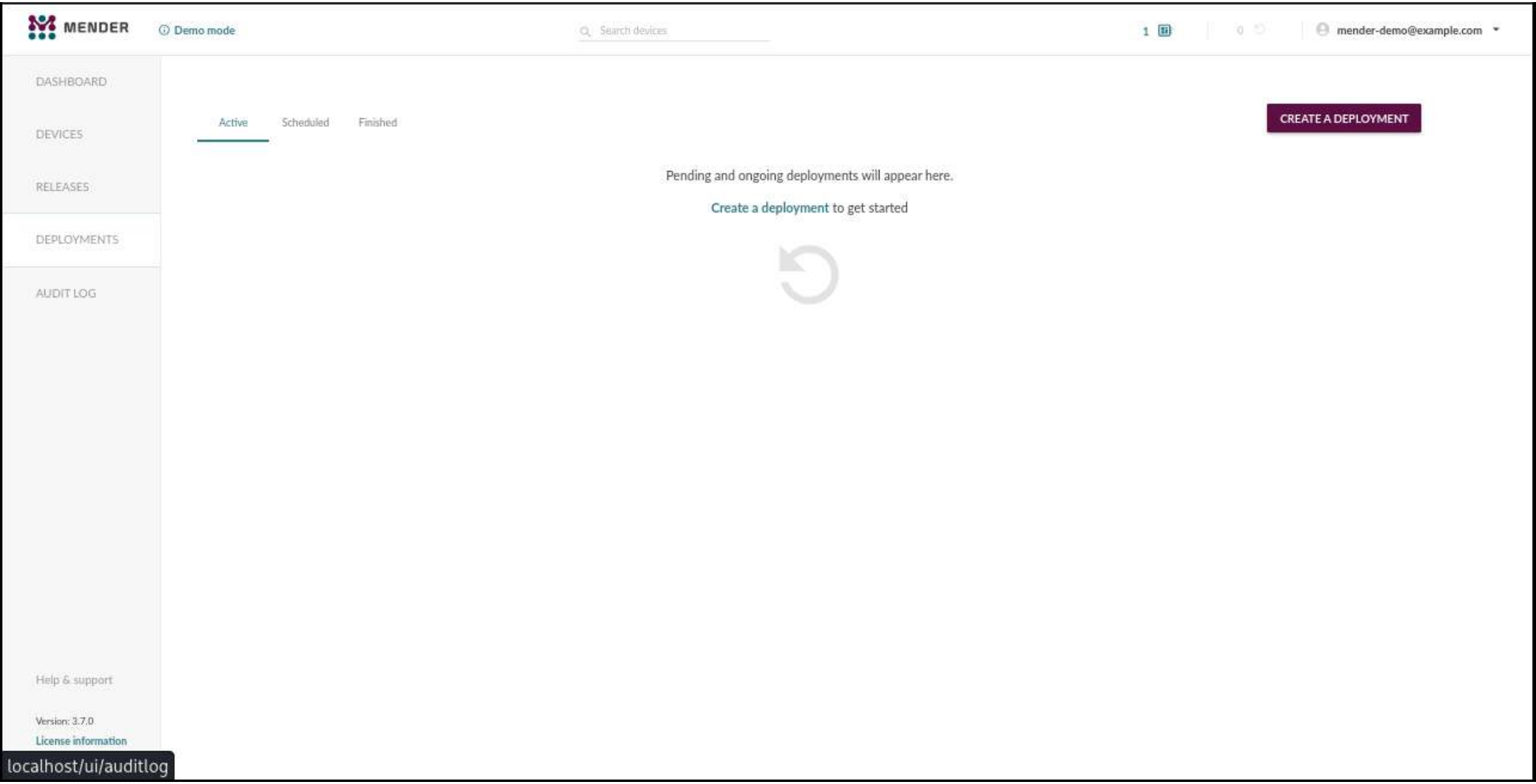

Mender
localhost/ui/releases
67%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
MENDER
Demo mode
Search devices
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Releases
Delta Artifacts generation
ENTERPRISE
UPLOAD
Release name
Starts with
Tags
Select tags
Contains Artifact type
Any
Name
Number of artifacts
Tags
Last modified
mender-demo-artifact-20260220-72fe482
1
-
2026-02-25 07:18
Rows 20
1-1 of 1
Help & support
Version: 3.7.0
License information
localhost/ui/deployments

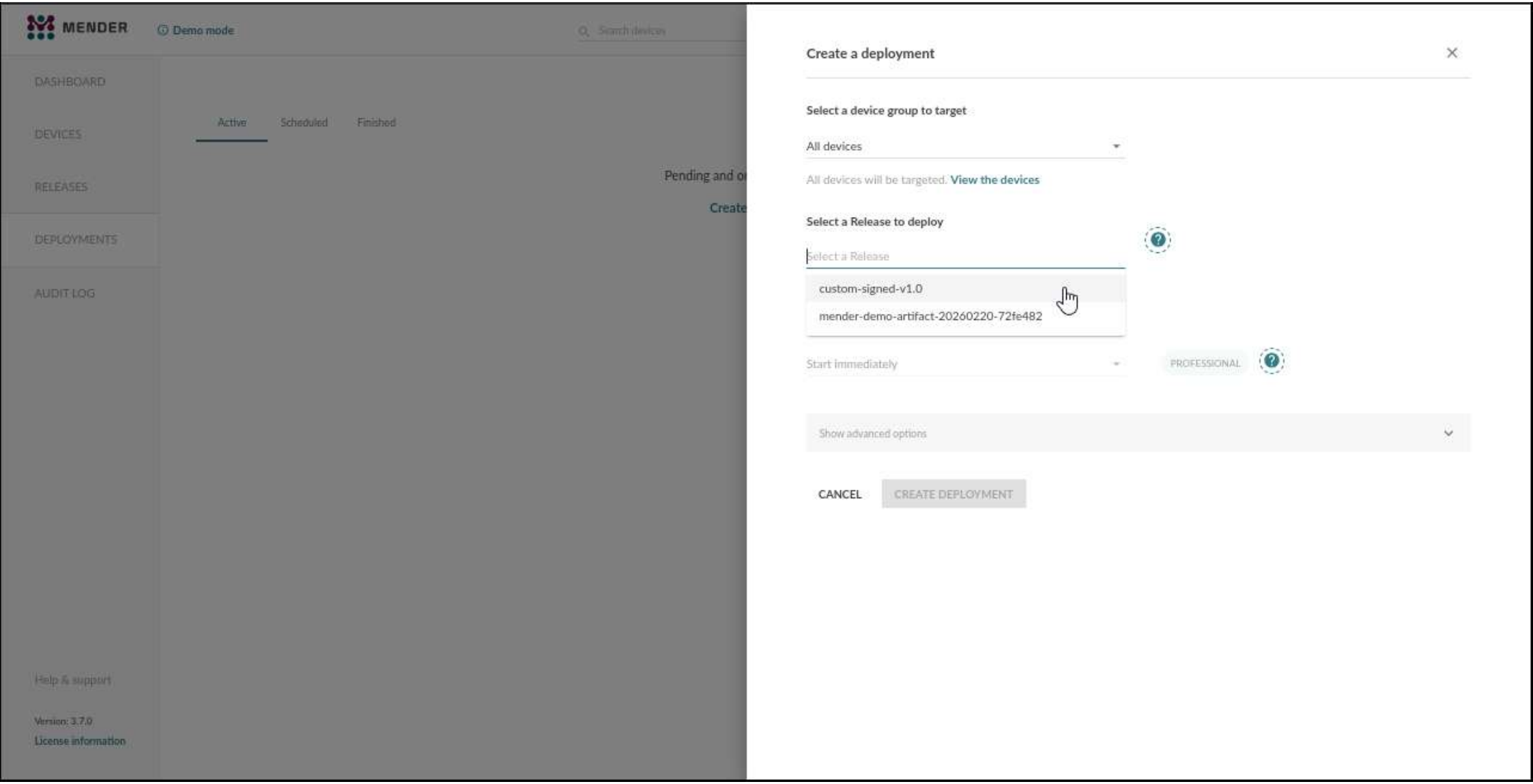

MENDER
Demo mode
Search devices
1
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Releases
Delta Artifacts generation
ENTERPRISE
UPLOAD
Release name
Tags
Starts with
Select tags
Name
Tags
Last modified
mender-demo-artifact-20260220-72fe482
-
2026-02-25 07:18
Rows 20
1-1 of 1
Help & support
Version: 3.7.0
License information
Upload an Artifact
Upload a premade Mender Artifact
OR
Upload a file to generate a single file application update Artifact
Drag and drop here or browse to upload
CANCEL

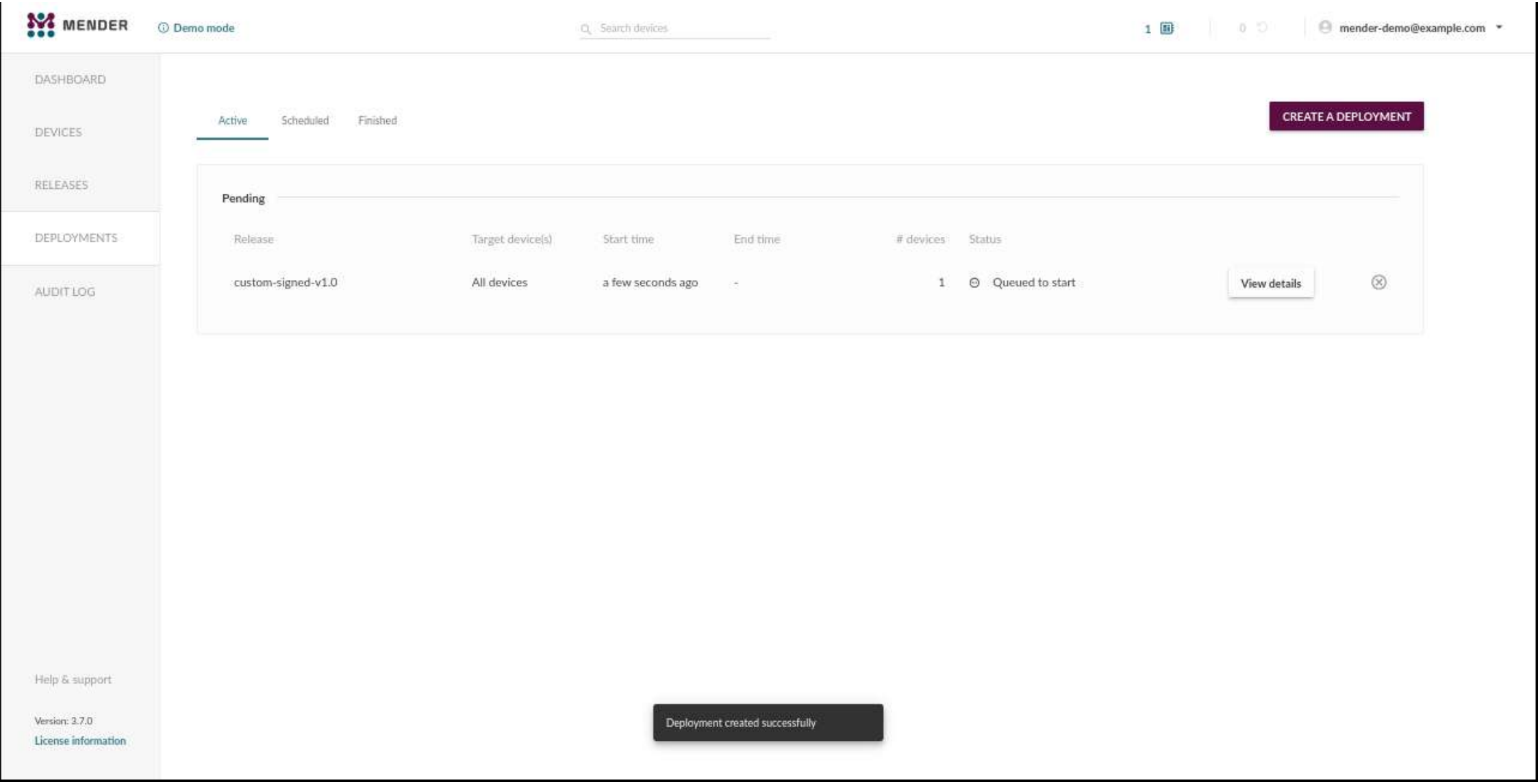

kali
mender-signing
Name
Size
Type
Modified
custom-signed-v1.mender
6.1 kB
Archive
05:52
legacy_private_key.pem
1.7 kB
Text
04:48
mender-artifact
29.6 MB
Program
15 Apr 2025
payload.sh
49 bytes
Program
04:49
payload.sig
256 bytes
Detached OpenPGP signature
04:50
private_key_pkcs8.pem
1.7 kB
Text
04:48
public_key.pem
451 bytes
Openssl PEM format
04:49

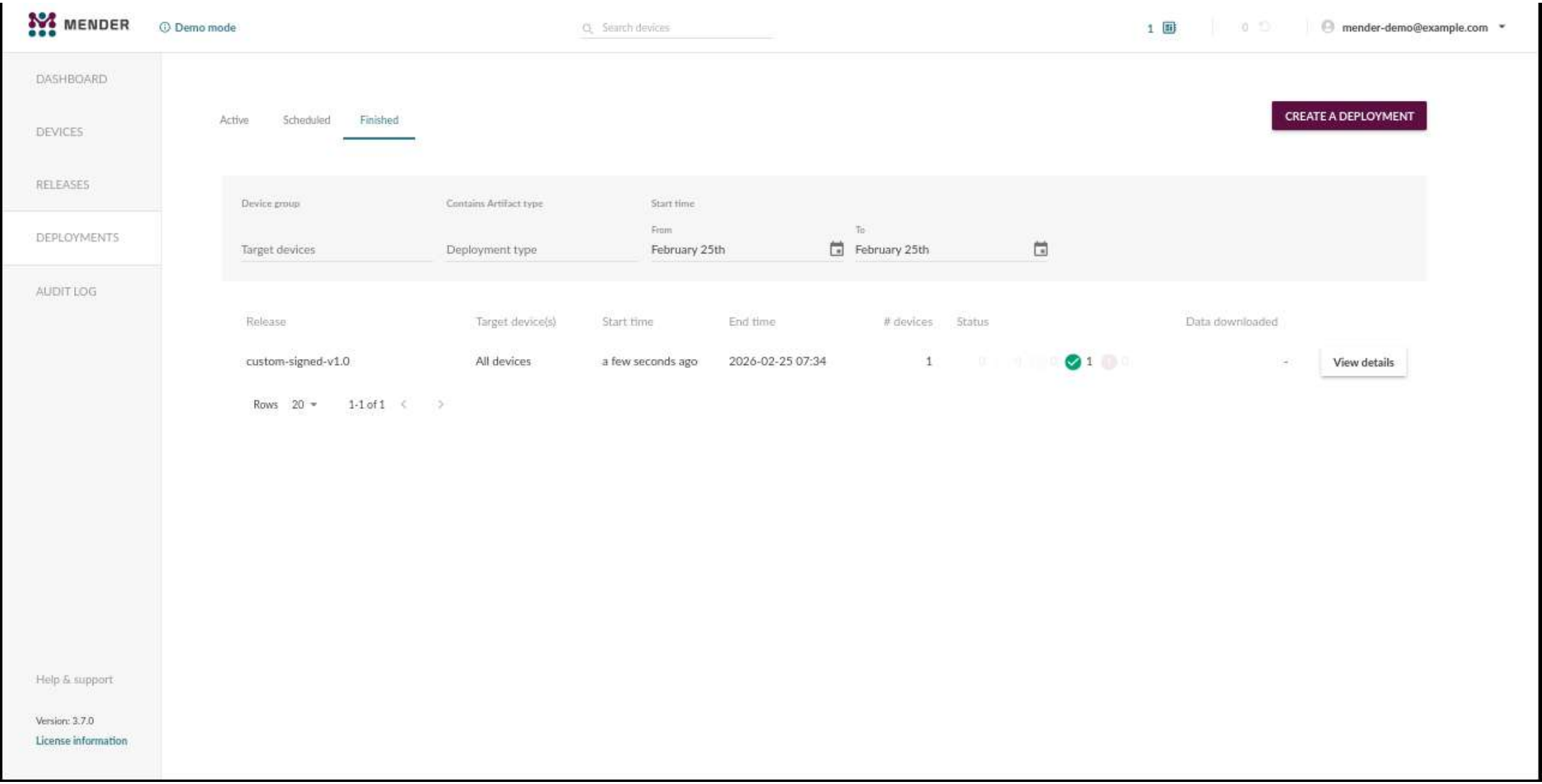

Upload an Artifact
Selected Mender Artifact
custom-signed-v1.mender
6.00 KB
CANCEL
UPLOAD ARTIFACT

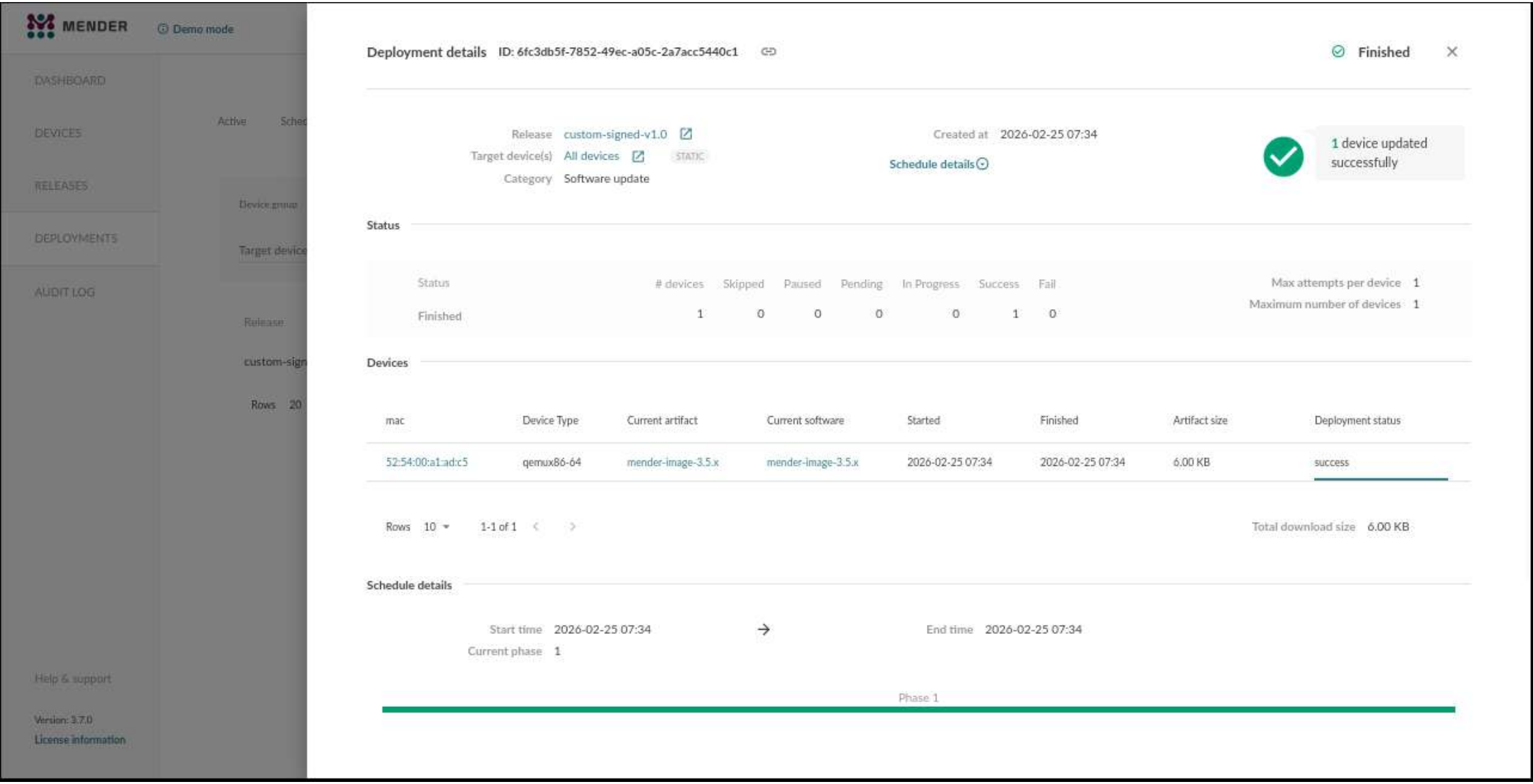

MENDER
Demo mode
Search devices
1
0
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Releases
Delta Artifacts generation
ENTERPRISE
UPLOAD
Release name
Starts with
Tags
Select tags
Contains Artifact type
Any
Name
Number of artifacts
Tags
Last modified
custom-signed-v1.0
1
-
a few seconds ago
mender-demo-artifact-20260220-72fe482
1
-
2026-02-25 07:18
Rows 20
1-2 of 2
Help & support
Version: 3.7.0
License information
Upload successful


MENDER
Demo mode
Search devices
1
0
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Active
Scheduled
Finished
CREATE A DEPLOYMENT
Pending and ongoing deployments will appear here.
Create a deployment to get started
Help & support
Version: 3.7.0
License information
localhost/ui/auditlog


MENDER
Demo mode
Search devices
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Active
Scheduled
Finished
Pending and o
Create
Help & support
Version: 3.7.0
License information
Create a deployment
Select a device group to target
All devices
All devices will be targeted. View the devices
Select a Release to deploy
Select a Release
custom-signed-v1.0
mender-demo-artifact-20260220-72fe482
Start immediately
PROFESSIONAL
Show advanced options
CANCEL
CREATE DEPLOYMENT


MENDER
Demo mode
Search devices
1
0
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Help & support
Version: 3.7.0
License information
Active
Scheduled
Finished
CREATE A DEPLOYMENT
Pending
Release
Target device(s)
Start time
End time
devices
Status
custom-signed-v1.0
All devices
a few seconds ago
-
1
Queued to start
View details
Deployment created successfully


MENDER
Demo mode
Search devices
1
0
mender-demo@example.com
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Help & support
Version: 3.7.0
License information
Active
Scheduled
Finished
CREATE A DEPLOYMENT
Device group
Target devices
Contains Artifact type
Deployment type
Start time
From
February 25th
To
February 25th
Release
Target device(s)
Start time
End time
devices
Status
Data downloaded
custom-signed-v1.0
All devices
a few seconds ago
2026-02-25 07:34
1
1
-
View details
Rows 20
1-1 of 1


MENDER
Demo mode
DASHBOARD
DEVICES
RELEASES
DEPLOYMENTS
AUDIT LOG
Help & support
Version: 3.7.0
License information
Deployment details ID: 6fc3db5f-7852-49ec-a05c-2a7acc5440c1
Finished
Release custom-signed-v1.0
Target device(s) All devices STATIC
Category Software update
Created at 2026-02-25 07:34
Schedule details
1 device updated successfully
Status
Status # devices Skipped Paused Pending In Progress Success Fail
Finished 1 0 0 0 0 1 0
Max attempts per device 1
Maximum number of devices 1
Devices
mac Device Type Current artifact Current software Started Finished Artifact size Deployment status
52:54:00:a1:ad:c5 qemux86-64 mender-image-3.5.x mender-image-3.5.x 2026-02-25 07:34 2026-02-25 07:34 6.00 KB success
Rows 10 1-1 of 1
Total download size 6.00 KB
Schedule details
Start time 2026-02-25 07:34
End time 2026-02-25 07:34
Current phase 1
Phase 1